

Multiobjective Differential Evolution for Higher-Dimensional Multimodal Multiobjective Optimization

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Abstract—In multimodal multiobjective optimization problems (MMOPs), there are several Pareto optimal solutions corresponding to the identical objective vector. This paper proposes a new differential evolution algorithm to solve MMOPs with higher-dimensional decision variables. Due to the increase in the dimensions of decision variables in real-world MMOPs, it is difficult for current multimodal multiobjective optimization evolutionary algorithms (MMOEA) to find multiple Pareto optimal solutions. The proposed algorithm adopts a dual-population framework and an improved environmental selection method. It utilizes a convergence archive to help the first population improve the quality of solutions. The improved environmental selection method enables the other population to search the remaining decision space and reserve more Pareto optimal solutions through the information of the first population. The combination of these two strategies helps to effectively balance and enhance convergence and diversity performance. In addition, to study the performance of the proposed algorithm, a novel set of multimodal multiobjective optimization test functions with extensible decision variables is designed. The proposed MMOEA is certified to be effective through comparison with six state-of-the-art MMOEAs on the test functions.

Index Terms—Benchmark functions, diversity measure, evolutionary algorithms, multimodal multiobjective optimization.

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I. INTRODUCTION

IN many practical applications, multiobjective optimization problems (MOPs) are quite normal. They contain multiple conflicting objective functions that need to be optimized simultaneously. In a general way, the formula of a MOP can be expressed as follows:

$$\begin{aligned} &\text{Minimize } \vec{F}(\vec{x}) = \{f_1(\vec{x}), f_2(\vec{x}), \dots, f_M(\vec{x})\}^T \\ &\text{s.t. } \vec{x} = (x_1, x_2, \dots, x_D) \in \mathbb{S} \end{aligned} \quad (1)$$

where $\vec{x}$ is a vector consisting of D decision variables in the decision space $\mathbb{S}$. $\mathbb{S}$ consists of all the possible values of $\vec{x}$, where $\vec{x} \in \{x_i^{\text{lower}} \leq x_i \leq x_i^{\text{upper}}, i = 1, \dots, D\}$, in which x_i^{lower} and x_i^{upper} respectively denote the lower bound and upper bound of x_i . $\vec{F}(\vec{x})$ is a vector consisting of M objective variables and objective space are composed of all the possible values of $f_j(\vec{x})$, where $j = 1, 2, \dots, M$.

In MOPs, a single solution is incapable of simultaneously optimizing all the objective functions due to the conflicts among multiple objectives. Different solutions are able to be compared by the Pareto dominance principle [1]: a solution $\vec{x}_a$ is deemed to be dominated by a solution $\vec{x}_b$, if $f_j(\vec{x}_b) \leq f_j(\vec{x}_a)$, and $\exists i = 1, 2, \dots, M, f_i(\vec{x}_b) < f_i(\vec{x}_a)$. If a solution is not Pareto dominated by any solution, it is said to be a non-dominated solution. Therefore, a set of non-dominated solutions named the Pareto optimal solution set (PS) is usually provided. The set of all the objective vectors corresponding to the PS is termed the Pareto front (PF).

In practical multiobjective optimization problems, there also exists multimodal multiobjective optimization problems (MMOPs), meaning there are multiple Pareto optimal solutions with equivalent objective values, as shown in Fig. 1. MMOPs can be found in many practical optimization problems, such as engine designing [2], [3], path planning [4], [5], feature selection [6]–[8], multi-project scheduling problems [9], ingredients problems [10], and protein structure prediction [11]. The primary objective in addressing MMOPs is twofold: acquiring a set of well-distributed objective vectors that approximate the true PF and maintaining their respective multiple equivalent Pareto optimal solution sets. However, current multiobjective optimization evolutionary algorithms (MOEAs) tend to overlook the distribution of solutions in the decision space. This oversight hinders the maintenance of diversity in the decision space throughout the population evo-

lution process, making it challenging to retain multiple equivalent PSs. Consequently, it becomes imperative to design MMOEAs specifically tailored to address these challenges.

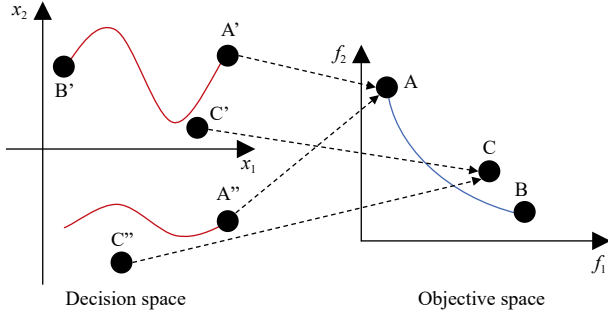


Fig. 1. Schematic diagram of multimodal multiobjective optimization problems.

Moreover, in some practical MMOPs, higher-dimensional decision variables may be involved, and they exhibit complex coupling relationships. These problems also possess highly nonlinear objective functions, making traditional methods difficult to solve. Specifically, within the context of the parameter optimization for cascade load force control design in parallel robot platforms [12]–[14], the cascade control parameters span 24 dimensions. In [14], an optimally tuned cascade controller utilizing the particle swarm optimization (PSO) algorithm [15] is introduced, offering a distinct advantage in addressing the formulated tracking problem. Beyond these characteristics, the motor design problem [3] presents MMOPs, where distinct design configurations can yield comparable levels of efficiency and reliability.

However, limited research exists on MMOPs with higher-dimensional decision variables, and current MMOEAs struggle to solve such optimization problems. While some existing MMO test functions can be extended to have multiple Pareto optimal solutions, these solutions often differ only in a few dimensions of the objective vector. In such cases, the decision variables can be optimized independently as their interdependencies are weak. As the dimensionality of decision variables increases, conventional methods like the crowding distance [16]–[19] or other approaches [6], [20]–[22] may result in inaccurate solution density estimates. Consequently, algorithms may fail to effectively maintain a diverse set of solutions during the optimization process. Therefore, it is needed to design specific methods for this situation [23].

Existing MMOEAs primarily focus on maintaining diversity in the decision space to preserve multiple equivalent Pareto solutions. However, their convergence performance is insufficient for effectively discovering high-quality solutions. Meanwhile, there are currently many large-scale processing techniques to effectively solve MOPs involving high-dimensional decision variables. Examples include decision variable grouping techniques [24]–[26], which analyze the correlation between decision variables to divide them into multiple groups and solve them systematically; decision space reduction techniques, which reduce the dimensionality of the decision space to narrow the search space and further reduce search difficulty [27]–[29]; and new search strategies [30]–

[32], which enhance the search capability of MOEAs to address the original problems. These MOEAs face challenges in preserving multiple equivalent Pareto optimal solutions due to the lack of operations that retain diversity and dominated solutions.

To address this limitation, this paper introduces a higher-dimensional multimodal multiobjective optimization differential evolution algorithm (HDMODE). In HDMODE, the original population is divided into two subpopulations, each serving a specific purpose. To handle MMOPs with increased search difficulty, the first subpopulation utilizes a convergence external archive and convergence preference evolutionary strategies to enhance the quality of final solutions. Meanwhile, the second subpopulation explores the remaining decision space using penalty strategies and an improved environmental selection method, further enhancing the search capability of the population and diversity performance. The improved environmental selection method introduces a novel approach to accurately estimate solution density in higher-dimensional decision space. This combination of strategies not only alleviates performance deterioration with increasing decision space dimensionality but also effectively maintains diversity to retain multiple Pareto solutions. Our research makes three key contributions:

- 1) A new framework of MMO test functions is proposed, incorporating various components with different functions to control convergence difficulty, multimodal characteristics, and the extensibility of decision variables. Based on this framework, fifteen test instances are generated, allowing for a more comprehensive evaluation of the performance of MMOEAs.

- 2) A dual-population coevolution method is designed to significantly enhance search capability. One subpopulation prevents performance deterioration by adopting a diversity quality assessment strategy and convergence archive, while the other subpopulation explores the remaining decision space. This method can effectively expand the search range of the population.

- 3) An improved environmental selection method is developed to preserve multiple Pareto optimal solutions. The proposed density estimation strategy adaptively evaluates the density of each solution in the decision space, enhancing the diversity performance of the population. Furthermore, this method can help the population to retain multiple equivalent Pareto optimal solutions with large differences.

The rest of the paper is structured as follows: Section II provides a review of related studies on MMOEAs and MMO test functions. Section III thoroughly illustrates the design principle, characteristics, and instances of the proposed test functions. In Section IV, HDMODE is presented and analyzed in terms of its principles and flow. A comparison experiment with other competitive MMOEAs and ablation experiments are described in Section V. Finally, Section VI concludes the paper and provides future research directions.

II. RELATED WORKS

A. Multimodal Multiobjective Optimization Benchmark Test Suits

To thoroughly investigate the performance of MMOEAs,

TABLE I
PROPERTIES OF CURRENT MULTIMODAL MULTIOBJECTIVE OPTIMIZATION TEST FUNCTIONS

Test suites	Scalable number of variables	Scalable number of variables that differ between different PSs	Scalable number of objectives	Scalable number of PSs	PF geometry	Coexistence of global and local PSs	Type of variables
TWO-ON-ONE [33]	√	×	×	×	Convex	×	Continuous
SYM-PART [34]	×	×	×	×	Convex	×	Continuous
Omni-test [35]	√	×	×	√	Convex	×	Continuous
Polygon [36]	×	×	√	√	Convex	×	Continuous
MMF [37]	×	×	×	×	Both	×	Continuous
Extended MMF [38]	√	×	√	√	Both	√	Continuous
MMMOP [39]	√	×	×	×	Concave	×	Continuous
IDMP [20]	×	×	×	×	Convex	×	Continuous
IDMP_e [40]	×	×	×	√	Convex	√	Continuous
SMMOP [41]	√	×	√	√	Convex	√	Discrete
Feature selection [6]	√	×	×	×	Convex	√	Discrete
Path planning [5]	×	×	×	×	Convex	×	Discrete

numerous MMO test suites have been designed. Initially, Preuss *et al.* [33] proposed a two-objective test function called TWO-ON-ONE, which features two equivalent Pareto optimal solution sets. The same authors later introduced the SYM-PART test function and its two variants, where the positions and shapes of Pareto sets are modified through rotation and translation [34]. Deb and Tiwari [35] developed the Omni-test problems, which allow for an adjustable number of Pareto sets. Ishibuchi *et al.* [36] designed Polygon-based problems in two-dimensional decision spaces, featuring multiple identical polygons with the objective of optimizing distances to all vertices. Each polygon contains a Pareto set, and the number of polygons corresponds to the number of Pareto sets. Subsequently, Liu *et al.* [42] expanded on the polygon-based problems by incorporating local Pareto sets and addressing the challenge of unbalanced diversity in both objective and decision spaces.

While these MMO test functions are relatively simple and easy to solve, subsequent researchers have designed benchmark test sets with more diverse properties. Yue *et al.* [37] introduced the MMF test suite, which includes discrete, non-linear Pareto fronts and Pareto sets, albeit with limited decision space dimensions and quantity of Pareto sets. They further added test functions with local Pareto sets and devised a method to visualize their landscapes [38]. These test functions exhibit relatively small search difficulties among different Pareto sets due to the limited decision space. Building upon multimodal optimization test functions [43], [44] and multi-objective optimization test functions [45], [46], Liu *et al.* [39] introduced MMMOPs, a scalable MMO test suite with seven decision variables and simple Pareto front shapes.

Although the existing test suites provide the same searching difficulties for multiple Pareto sets, different Pareto sets may encounter distinct challenges in practical optimization problems. Recognizing this, Liu *et al.* [20] introduced the concept of imbalanced difficulties in convergence and diversity in the decision space, and designed the IDMP test suite, where the difficulties of searching different Pareto sets can be adjusted.

Li *et al.* [40] incorporated additional multimodal test functions based on IDMP to construct local Pareto sets and Pareto fronts. Considering that many practical optimization problems are discretely coded, several discrete and other MMO test functions have been proposed. Yue *et al.* [6] identified multimodal properties in multi-objective feature selection problems. Tian *et al.* [41] devised large-scale sparse test functions based on the design optimization problem of convolutional neural networks (CNNs). Liang *et al.* [5] organized an MMO path planning competition that included three types of problems for testing. The properties of these MMO test functions are summarized in Table I.

However, most existing MMO test suites exhibit low convergence difficulty in finding the true Pareto front, and the dimensions of decision variables related to multimodality are also limited. This limitation hinders the comprehensive evaluation of MMOEAs. Therefore, there is a need to design higher-dimensional and more challenging MMO test functions to encompass these missing characteristics.

B. Multimodal Multiobjective Optimization Evolutionary Algorithms

MMOEAs face the dual challenge of acquiring well-distributed objective vectors approximating the true PF in the objective space and ensuring sufficient diversity in the decision space to preserve multiple Pareto sets. Deb and Tiwari [35] previously introduced a nearest-neighbor-based tournament for parent selection and introduced the concept of crowding distance in the decision space.

A prevalent strategy involves employing MOEAs as search engines and integrating them with various niching mechanisms to safeguard distant yet promising solutions from elimination. For example, DN-NSGA-II [47] adopted a novel niching method in mating selection to generate more diverse solutions. Yue *et al.* [37] introduced a ring topology to reduce information interaction between solutions, inducing multiple niches. They also modified the crowding distances of boundary solutions to mitigate their negative effects. Liu *et al.* [42]

implemented a sharing-based niching method in both decision and objective spaces, enhancing the selecting probability of well-distributed solutions. Qu *et al.* [48] introduced a self-organizing speciation strategy in the multi-objective PSO algorithm, improving population diversity by selecting seeds based on non-dominated sorting value and gradually generating niches according to the preset neighborhood radius. In TriMOEA-TA&R [39], the decision variable analysis technique separated all decision variables into two archives, utilizing a clustering strategy and clearing-based niching methods to ensure diversity performance in both spaces. Tanabe and Ishibuchi [22] introduced addition and deletion operators to MOEA/D [49], enabling its reference vectors to preserve multiple solutions through niching methods. Yue *et al.* [18] considered the non-dominated sorting value when calculating the SCD value of the solution, further enhancing the diversity of the updated population.

In addition to the traditional niche methods mentioned above, various clustering methods have been employed in different parts of MOEAs to maintain population diversity during the evolutionary process. Lin *et al.* [50] proposed a dual clustering strategy to preserve global and local Pareto sets. Initially, a neighborhood-based clustering method groups nearby solutions in the decision space, followed by hierarchical clustering to ensure a well-distributed distribution of objective vectors. Liang *et al.* [51] designed a distance-based elite solution learning strategy to help the population learn from non-dominated solutions with good diversity. They also proposed a clustering-based special crowding distance strategy to improve the accuracy of diversity measurement. Zhou *et al.* [19] used the affinity propagation (AP) clustering method to selectively retain non-dominated solutions in the cluster, prioritizing solutions with good diversity in the decision space.

Another effective approach involves evaluating solutions by integrating their density in both spaces and their convergence qualities. Liu *et al.* [20] proposed a solution assessment method called CPDEA, improving search efficiency by considering the mutual influence of local convergence quality. The k -nearest neighbor method is applied in both spaces. Similarly, MMEA-WI [21] introduced a convergence archive and a weighted indicator to evaluate solutions. The weighted indicator is calculated by summing fitness values based on their distances. The same authors [40] employed a method for evaluating local convergence quality and a hierarchical ranking strategy to solve MMOPs with local Pareto sets.

Furthermore, several MMOEAs have been developed to address discrete, sparse, and constrained MMOPs. For instance, Tian *et al.* [41] measured similarities among different subpopulations and utilize a guiding vector method to differentiate the search directions of the sub-populations to solve large-scale sparse MMOPs. Subsequently, Ding *et al.* [52] proposed a hybrid hierarchical clustering method to divide sub-populations and designed a global and local guided information fusion strategy to improve the convergence of the population in large-scale sparse search spaces. Peng and Ishibuchi [53] proposed an MMOEA based on MOEA/D [49], wherein each weight vector is associated with a subpopula-

tion. To sustain diversity in the decision space when addressing large-scale MMOPs, they incorporated both a clearing mechanism and a greedy removal mechanism. Additionally, Liang *et al.* [54] designed an MMOEA incorporating a species conservation mechanism to generate multiple subpopulations. They combined the species conservation mechanism and a constrained handling technique to explore feasible regions in the decision space, furthering the resolution of MMOPs with constraints.

The increasing dimensionality of decision variables presents a greater challenge to algorithms, and current MMOEAs may suffer from serious performance degradation due to the lack of a specific mechanism. To address this issue, a new MMOEA is proposed in Section IV.

III. PROPOSED MULTIMODAL MULTIOBJECTIVE OPTIMIZATION BENCHMARK SUITS

To facilitate the study of MMOPs and overcome the limitations of current benchmarks, a new set of test functions called HDMMFs is introduced. These HDMMFs are designed based on the bottom-up design principle of multiobjective optimization benchmarks [55]. They inherit the fundamental properties of existing MMO benchmarks, such as multiple equivalent Pareto sets, known true Pareto sets and Pareto fronts, and the ability to scale the number of objective functions and decision variables. However, the proposed HDMMFs have been modified to better simulate real-world problem characteristics. The modifications include:

- 1) Enhancing the differences between different equivalent Pareto solutions.
- 2) Allowing for flexible grouping of decision variables to capture complex relationships.
- 3) Introducing a mapping relation in the polar coordinate system to incorporate different solution patterns.
- 4) Proposing hybrid linkages that couple different groups of variables to capture their interdependencies.
- 5) Increasing the relevance of decision variable dimensions to the position of Pareto solution sets and the shape function.

These modifications aim to create HDMMFs that more accurately represent the complexities and challenges encountered in practical optimization problems. By incorporating these design enhancements, the HDMMFs provide a more comprehensive and realistic evaluation of the performance of MMOEAs.

A. Notations Used in the Formulation

In HDMMF, the decision variables of each solution are divided into three parts, corresponding to three component functions respectively. Specifically, a solution with D decision variables is divided into the shape component function related variables $(x_1, \dots, x_m)$, the multimodal component function related variables $(x_{m+1}, \dots, x_{m+k})$, the distance component function related variables $(x_{m+k+1}, \dots, x_D)$. Furthermore, the polar vector $\vec{\theta}$ is composed by $(\theta_1, \dots, \theta_{m-1})$, and the polar coordinate variables $(\theta_1, \dots, \theta_{m-1})$ are defined by the decision variables $(x_1, \dots, x_m)$, the specific formula is shown in (5). In order to avoid confusion, the definition summary of the parameters in HDMMF is shown in Table II.

TABLE II
NOTATION USED IN THE PROBLEM FORMULATION

Notation	Description
m	The number of objectives
D	The number of decision variables
$\mathbf{x}$	The solution containing D decision variables
$f_i(\mathbf{x})$	The i objective function value of $\mathbf{x}$
$\vec{\theta}$	A vector containing $m-1$ polar coordinate variables
θ_j	The polar coordinate variable θ transformed by the j th decision variable x_j
$g_i(\vec{\theta})$	The i th shape function value of $\vec{\theta}$
k	The number of decision variables for the multimodality component $q(x_k)$
$q(x_k)$	The function value of the decision variables $(x_{m+1}, \dots, x_{m+k})$
l	The number of decision variables for the distance function $h(x_l)$
$h(x_l)$	The function value of the decision variables $(x_{m+k+1}, \dots, x_D)$

B. Basic Formulation

In typical MMO test problems, equivalent PSs are often constructed by applying translation and rotation operations, which can lead to multiple PSs with similar shapes and fitness landscapes. However, in practical optimization problems, most of these equivalent optimal solutions are irrelevant and do not accurately represent the problem characteristics. Therefore, it is essential to establish distinct mapping relationships between multiple equivalent PSs and the objective function. To address this issue, this paper proposes a new benchmark test set that aims to capture the complexities and diversities of practical optimization problems. The basic formula of the proposed benchmark test set is as follows:

$$\min \begin{cases} f_1(\mathbf{x}) = g_1(\vec{\theta})(1 + p(\mathbf{x}^m) + q(\mathbf{x}^k) + h(\mathbf{x}^l)) \\ \dots \\ f_m(\mathbf{x}) = g_m(\vec{\theta})(1 + p(\mathbf{x}^m) + q(\mathbf{x}^k) + h(\mathbf{x}^l)) \end{cases} \quad (2)$$

where $g(\vec{\theta})$ is the shape function to construct different PF shapes, which may bring different convergence difficulties in the objective space [56]. $p(\mathbf{x}^m)$ is related to the first m decision variables, which together with $g(\vec{\theta})$ determines the diversity difficulty of the problem. The function $q(\mathbf{x}^k)$ is related to the subsequent k decision variables and can be used to test the ability to retain equivalent PSs. The distance function $h(\mathbf{x}^l)$ can be used to extend the decision variables, and it controls the convergence difficulty together with $q(\mathbf{x}^k)$. These three component functions $p(\mathbf{x}^m)$, $q(\mathbf{x}^k)$, and $h(\mathbf{x}^l)$ transform their theoretical minimum to zero through operations such as shifting and deformation (e.g., squaring and taking absolute values).

C. Shape Component

The shape of the true PF is also an important problem to be considered during the construction of MMOP test functions, because different shapes will lead to different diversity difficulties [56]. In this paper, the shape function is employed to govern the shape of the true PF, incorporating five fundamental characteristics: linear, concave, convex, mixed, and dis-

cretely irregular. The detailed formulation of $g(\vec{\theta})$ for generating concave PF is shown as an example,

$$g_i(\vec{\theta}) = \begin{cases} \cos\left(\frac{\pi}{2}\theta_1\right), & i = 1 \\ \prod_{j=1}^{i-1} \sin\left(\frac{\pi}{2}\theta_j\right) \cos\left(\frac{\pi}{2}\theta_i\right), & i = 2, \dots, m-1 \\ \prod_{j=1}^{i-1} \sin\left(\frac{\pi}{2}\theta_j\right), & i = m. \end{cases} \quad (3)$$

When all the Pareto optimal solutions are found, that is, all three functions $p(\mathbf{x}^m)$, $q(\mathbf{x}^k)$, and $h(\mathbf{x}^l)$ obtain the global minimum of zero. The formula of the true PF can be extrapolated as follows:

$$\sum_{i=1}^m f_i(\mathbf{x})^2 = \sum_{i=1}^m g_i(\vec{\theta})^2 = 1, \quad i = 1, \dots, m \quad (4)$$

where $\vec{\theta}$ is a vector $(\theta_1, \theta_2, \dots, \theta_{m-1})$ in connection with the first m dimensions of the decision variables. It determines the mapping relationship between the decision variable and the objective vector, and we can construct various equivalent PSs by constructing different mapping relationships, which is explained in detail in the next subsection. Its basic formulation is provided as follows:

$$\theta_j = \frac{2}{\pi} \arctan \frac{\sqrt{\sum_{d=j+1}^m x_d^2}}{x_j}, \quad j = 1, \dots, m-1. \quad (5)$$

The above steps constitute a basic multiobjective optimization benchmark test set framework, which contains multiple flexible components to locally control different characteristics.

D. Multimodality Component

The third function $q(\mathbf{x}^k)$ is related to the next k decision variables $(x_{m+1}, x_{m+2}, \dots, x_{m+k})$. This function is mainly utilized to improve the diversity difficulty by controlling the differences between multiple equivalent PSs and the number of PSs. An easy example is shown as follows:

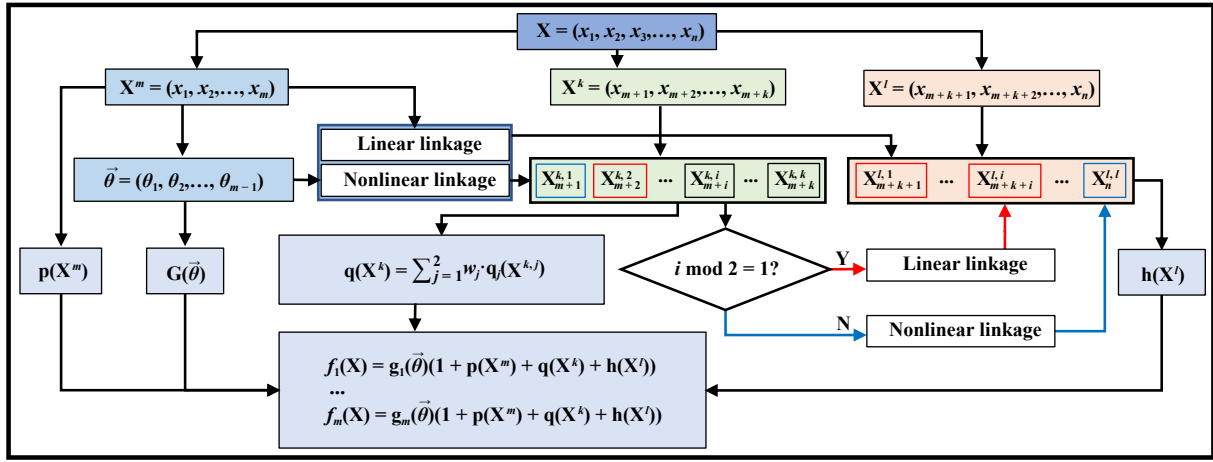


Fig. 2. HDMMF test function construction flowchart.

$$\begin{cases} q(\mathbf{x}^k) = \gamma_1 \left(1 - \sin\left(\frac{\pi}{n}(t_i - \beta)\right) \right)^2 + \gamma_2 \sum_{i=2}^k i(\omega t_i + t_{i-1})^2 \\ t_i = \sum_{j=m+1}^i x_j, \quad i = m+1, \dots, m+k \end{cases} \quad (6)$$

where the parameters β , ω , and n are used to control the positions and number of multiple equivalent PSs. According to (6), when $q(\mathbf{x}^k)$ is the minimum value, t_i is equal to $n/2 + \beta$, and β controls the value of t_i . Meanwhile, t_{i-1} is equal to $-t_i/\omega$, ω controls the relation between t_i and t_{i-1} . γ controls the weight coefficient of each subfunction on $q(\mathbf{x}^k)$ and it influences the degree of objective value changes caused by the perturbation of solutions. Then, the value of each decision variable dimension x_j can be further deduced.

In this way, the number of multiple equivalent PSs in this test function and the dimensions of the decision variables with differences between PSs are extensible, not limited to two or three dimensions. In addition, multiple current multimodal functions can be combined and embedded into this framework to improve the solving difficulty and the position of the Pareto optimal solution changes as the decision variables increase. Detailed steps are shown in Fig. 2, where different weights are applied to multimodal functions with different characteristics to construct $q(\mathbf{x}^k)$.

E. Distance Component and Linkage Functions

The distance function $h(\mathbf{x}^l)$ is employed to test the convergence performance of the true PF, which is concerned with the remaining dimensions. In this paper, the common single objective optimization test function is adopted to construct the distance function, which can be found in [57]–[59]. It can be seen from Fig. 2, where each part of the variable dimensions has flexibly coupling relationships with other parts through different linkage functions, which are defined in [60]. In this paper, the linkage functions are slightly improved and the detailed linear formulation $L(x_j^m, x_l)$ is provided as follows:

$$x_j^{m,i} = \left(1 + \frac{i}{|\mathbf{x}^m|} \right) (\alpha \cdot x_j^{m,i} - L_i) - \frac{x_l}{U_l - L_l} \cdot (U_i - L_i). \quad (7)$$

The nonlinear formulation is shown as follows:

$$\begin{aligned} x_j^{m,i} &= \left(1 + \cos(0.5\pi \frac{i}{|\mathbf{x}^m|}) \right) (\alpha \cdot x_j^{m,i} - L_i) \\ &\quad - \frac{x_l}{U_l - L_l} \cdot (U_i - L_i) \end{aligned} \quad (8)$$

where $x_j^{m,i}$ indicates the i -th dimension in $\mathbf{x}^m$, in which j means its actual order in the whole dimensions of decision variables. x_l is the other dimension that is used to link with $\mathbf{x}^m$. U and L denote respectively the maximum value and minimum value of the decision variables on each dimension. α is a parameter to ensure the transformed decision variables remain within the search boundary. In this way, the coupling relationships between different parts of the decision variables are controllable.

F. Instances and Characteristics of HDMMFs

The MMO test functions can be constructed by the above framework, in which the dimension of each part can be extended. Based on the above framework and design principle, fifteen HDMMFs are designed as instances for comparing the proposed algorithm and other competitive MMOEAs. The characteristics of these instances are listed in Table III-F. Next, the detailed characteristics of HDMMFs are summarized as follows:

- 1) The number of objectives, true PSs, dimension of decision variables, and differences among multiple PSs are controllable.
- 2) Various PFs with different shapes, such as linear, nonlinear, irregular, and discontinuous, exist in the objective space. The positions of multiple true PSs can change as the dimension of decision variables increases.
- 3) The HDMMFs have flexible component functions to control diversity and convergence difficulty, making them more capable of simulating real optimization problems compared to other MMO benchmarks.
- 4) The HDMMFs enhance the coupling relationships among the dimensions of decision variables by adopting improved linkage functions.
- 5) The mapping relationship from decision variables to objective vectors is more intuitive, facilitating the exploration of potential properties.

Due to these complex characteristics, the proposed bench-

TABLE III

THE PROPERTIES OF HDMMF, WHERE $D_{\min}$, $M_{\min}$ RESPECTIVELY DENOTE THE MINIMUM QUANTITY OF DECISION VARIABLES AND OBJECTIVE FUNCTIONS. N PRESENTS THE QUANTITY OF EQUIVALENT PSS. C_1 , C_2 , AND C_3 DENOTE THREE DIFFERENT PROPERTIES RESPECTIVELY

Test functions	$D_{\min}$	$M_{\min}$	N	C_1	C_2	C_3
HDMMF1	7	2	2	×	×	√
HDMMF2	10	2	Any	√	√	√
HDMMF3	10	2	4	√	√	√
HDMMF4	10	2	5	√	√	√
HDMMF5	16	2	4	√	√	√
HDMMF6	12	2	Any	√	√	√
HDMMF7	12	2	4	√	√	√
HDMMF8	12	2	Any	√	√	√
HDMMF9	25	2	3	√	√	√
HDMMF10	15	2	2	√	√	√
HDMMF11	18	2	Any	√	√	√
HDMMF12	18	2	Any	√	√	√
HDMMF13	60	3	2	√	√	√
HDMMF14	10	2	4	×	√	√
HDMMF15	30	2	Any	√	√	√

C_1 : Scalable number of objectives and variables

C_2 : Existence of coupling relationship between decision variables

C_3 : Scalable number of variables that differ between different PSS

mark presents significant challenges for current MMOEAs in finding pareto optimal solutions, further maintaining multiple PSSs. These challenges highlight the need for specific MMOEAs to effectively address the higher-dimensional MMOPs.

IV. PROPOSED METHOD

In this section, a new multiobjective differential evolution algorithm with a dual-population framework is proposed. Moreover, the detailed procedure is provided and its internal mechanisms have also been analyzed in detail.

A. Procedure of HDMMODE

The framework of HDMMODE is described in Algorithm 1 and the illustration of this framework is shown in Fig. 3, in which the algorithm starts with the initialization of two populations $\mathbf{P}_1$ and $\mathbf{P}_2$ with N solutions. The population $\mathbf{P}_1$ firstly generate offspring $\mathbf{Q}_1$ through the differential evolution (DE) operator [61], which is shown detail in Algorithm 2. All non-dominated solutions of the first population are saved in each generation to update the convergence archive $\mathbf{A}_1$. Then, a conditional judgment is made on $\mathbf{P}_1$, that is, if the difference between the normalized objective values of the population $\mathbf{P}_1$ and the previous population is less than the threshold, the population size begins to shrink through environmental selection. The population shrinks by one solution each generation until it has shrunk to half its original size. Meanwhile, the number of solutions in $\mathbf{P}_2$ is increased at the same time to ensure that the total number of solutions in each generation remains the same. Otherwise, it remains unchanged, as shown in lines 5–11. This condition is used to judge whether the population has fallen

into the local optimal region or has converged to the global optimal region. When the population is in this situation, it should be given reduced computing resources [62].

Algorithm 1 Framework of HDMMODE

Input: an MMOP with m objective functions and n decision variables

Output: the final population $\mathbf{P}$

```

1: Initialization:  $\mathbf{P}_1$  and  $\mathbf{P}_2$  with  $N$  solutions,  $FE_{\max}$ 
2:  $FE = 0$ ,  $N_d = N$ ,  $\mathbf{A}_1 = \emptyset$ ;
3: while  $FE \leq FE_{\max}$  do
4:    $\mathbf{U}_1 = \mathbf{P}_1 \cup \mathbf{A}_1$ 
5:    $\mathbf{Q}_1 = \text{Reproduction}(\mathbf{U}_1, \text{length}(\mathbf{U}_1), \text{true})$ 
6:   if  $\text{Conversion}(\mathbf{P}_1) = \text{True}$  or  $\text{Flag} = \text{True}$  then
7:      $\mathbf{P}_1 = \text{Environmental\_Selection}(\mathbf{Q}_1, \mathbf{P}_1, N_d)$ 
8:      $\text{Flag} = \text{True}$ 
9:      $N_d = N_d - 1$ 
10:  else
11:     $\mathbf{P}_1 = \text{Environmental\_Selection}(\mathbf{Q}_1, \mathbf{P}_1, N)$ 
12:  end if
13:   $\mathbf{A}_1 = \text{Archive}(\mathbf{P}_1, \mathbf{A}_1)$ 
14:   $\mathbf{Q}_2 = \text{Reproduction}(\mathbf{P}_2, \text{length}(\mathbf{P}_2), \text{false})$ 
15:   $\text{near} = \text{Neighbor\_Identification}(\mathbf{P}_2, \mathbf{A}_1)$ 
16:   $\mathbf{P}_2(\text{near}) = \text{Penalty}(\mathbf{P}_2(\text{near}))$ 
17:   $\mathbf{P}_2 = \text{Environmental\_Selection}(\mathbf{Q}_2, \mathbf{P}_2, 2N - N_d)$ 
18:   $FE = FE + 2N$ 
19: end while

```

The method for updating $\mathbf{P}_2$ is different. After its reproduction operation, the solutions $\mathbf{P}_2(\text{near})$ in the neighbor range of $\mathbf{A}_1$ is selected. The range is a ratio α of the difference between the upper and lower bounds of current population on each dimension. Then a penalty function is imposed on the solutions $\mathbf{P}_2(\text{near})$ to reduce their convergence qualities, which can help the population $\mathbf{P}_2$ to explore other space, as shown in lines 12–14. The second population uses the reproduction method based on HAD distance to produce more diverse solutions, and when the first population reduction condition is triggered, the individual size of the second population is gradually increased to ensure that the total number of individuals in the population remains unchanged, as shown in lines 15–17. If the termination condition is satisfied, the final population is output.

B. Reproduction

This paper adopts three mutation strategies to generate offspring. These general formulas are as follows:

1) DE/current-to-best/1/bin

$$v_i = x_i + F \times (x_{\text{best}} - x_i) + F \times (x_{\text{rand1}} - x_{\text{rand2}}). \quad (9)$$

2) DE/current-to-better/1/bin

$$v_i = x_i + F \times (x_{\text{better}} - x_i) + F \times (x_{\text{rand1}} - x_{\text{rand2}}) \quad (10)$$

$$u_{i,j} = \begin{cases} v_{i,j}, & \text{if } \text{rand}_j < \text{CR} \text{ or } j = j_{\text{rand}} \\ x_{i,j}, & \text{otherwise.} \end{cases} \quad (11)$$

3) DE/current-to-rand/1

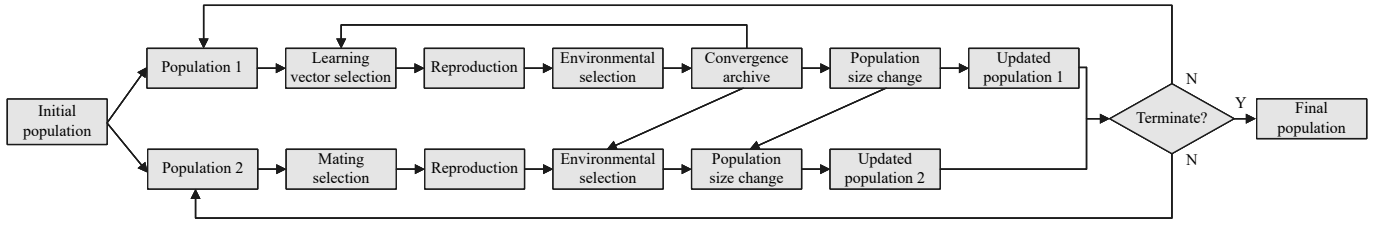


Fig. 3. The framework flow chart of HDMODE.

$$u_i = x_i + rand \times (x_{rand1} - x_i) + F \times (x_{rand2} - x_{rand3}) \quad (12)$$

where x_{best} refers to a nondominated solution that is randomly selected in an alliance of all non-dominated solutions in the convergence archive and current population. The selection of x_{best} as the learning vector aims to enhance the convergence quality of the generated solutions and further improve the search ability of the population. In addition, x_{better} is a solution randomly selected with lower Pareto dominance rank and higher D value than x_i . x_{better} serves as the learning vector aimed at enhancing the diversity quality of generated solutions.

The detailed process of reproduction is shown in Algorithm 2. The first population produces offspring through two mutation strategies that favor convergence performance, as shown in lines 2–9. The second population uses a tournament selection mechanism based on the HAD method [63] to select the parent and uses the third mutation strategy to produce a more diverse solution, as shown in lines 11–14. The formula for calculating HAD values is as follows:

$$HAD(x_i) = \frac{N-1}{\sum_{j=1, j \neq i}^N 1/ED(x_i, x_j)} \quad (13)$$

where $ED(x_i, x_j)$ represents the Euclidean distance between x_i and x_j . In addition, x_j represents any solution different from x_i in the population. With the above strategies, the first population can produce better solutions generation by generation, and the second population can select sparse solutions as the parent generation to help the population produce more diverse solutions.

Algorithm 2 Reproduction

Input: Population $\mathbf{P}$, N , $mask$ **Output:** Reproduced population $\mathbf{Q}$

```

1:  $\mathbf{Q} = []$ ;
2: if  $mask = \text{True}$ 
3:   for  $i = 1 : N$ 
4:     if  $rand < 0.5$ 
5:        $\mathbf{Q}(i) = \text{Operator}(\mathbf{P}(i))$  according to (9)–(11);
6:     else
7:        $\mathbf{Q}(i) = \text{Operator}(\mathbf{P}(i))$  according to (12);
8:     end if
9:   end for
10: else
11:   Calculate the HAD value according to (13);
12:    $Index = \text{Tournament\_selection}(\mathbf{P}, HAD)$ ;
13:    $\mathbf{Q} = \text{Operator}(\mathbf{P}(Index))$  according to (12);
14: end
15: Return  $\mathbf{Q}$ 

```

C. Environmental Selection

The general procedure of the proposed environmental selection mechanism is presented in Algorithm 3, in which N represents the number of new population $\mathbf{P}^*$. Firstly, all the decision variables and objective function values of the population $\mathbf{P}$ are normalized separately. The diversity quality of each solution is evaluated and the specific evaluation process is shown in Algorithm 4. All the solutions are sorted [1] and the neighborhood-based clustering method [50] is employed to divide the population into multiple clusters as shown in lines 2–6. In each cluster, the non-dominated solution with the highest diversity quality is reserved. The solutions of different Pareto fronts are successively selected according to their Pareto rankings, as shown in lines 7–12. The process will not stop until the number of solutions exceeds N . Then, the diversity qualities of the solutions on the last Pareto front F_i are assessed and the solutions with higher diversity qualities are selected successively until the total quantity reaches N .

Algorithm 3 Environmental Selection

Input: Population $\mathbf{P}$, Offspring $\mathbf{Q}$, Population Size N **Output:** New Population $\mathbf{P}^*$

```

1: Normalize  $\mathbf{P}$  in decision and objective space
2:  $D = \text{Diversity Quality Calculation}(\mathbf{P}, N)$  // Algorithm 4
3:  $(F_1, \dots, F_n) = \text{Non-dominated Sorting}(\mathbf{P}, \mathbf{Q})$ 
4:  $(C_1, \dots, C_k) = \text{Neighborhood-based Clustering}(\mathbf{P}, \mathbf{Q})$ 
5:  $\mathbf{P}^* = \text{The non-dominated solutions with the highest } D \text{ value in each cluster } (C_1, \dots, C_k)$ 
6:  $i = 1$ 
7: while  $|\mathbf{P}^*| < N$ 
8:    $\mathbf{P}^* = \mathbf{P}^* \cup \mathbf{P}(F_i)$ 
9:    $i = i + 1$ 
10: end while
11:  $N_r = N - |\mathbf{P}^*|$ 
12: if  $N_r \geq 0$ 
13:   for  $j = 1 : \text{length}(F_{i+1})$ 
14:      $D(j) = \text{Diversity Quality Calculation}(\mathbf{P}(F_{i+1}), \text{length}(F_{i+1}))$ 
15:   end for
16:    $Index = \text{Decending Sorting}(D)$ 
17:    $\mathbf{P}^* = \mathbf{P}^* \cup \mathbf{P}(F_{i+1}(Index(1 : N_r)))$ 
18: end if
19: Return  $\mathbf{P}^*$ 

```

The pseudocode of the diversity quality assessment method is in Algorithm 4. Firstly, the crowding distances of all solutions in the objective space are calculated. The Euclidean distances between each solution and the others are calculated and the population is sorted in ascending order according to the distances. Then, the nearest k solutions are selected as the ini-

tial neighborhood solutions and the mean and standard deviation values of their decision variables at each dimension are calculated, as shown in lines 2–6. Then the neighborhood range of each solution is fitted according to the known mean values, standard deviation values, and the three-sigma principle in the Gaussian distribution [64], [65]. If the nearest $(k+1)$ th solution lies in the niche range, it is included in this niche and the niche range is refitted by updating the mean and standard deviation values. This process does not stop until no solution lies in the niche range. Through this process, a niche is fitted for each solution and the density of each solution D_x in the decision space is estimated by calculating the product of the mean and minimum of the distance between each solution and all solutions in its niche, as shown in lines 7–16. This D_x value can comprehensively represent the crowding degree of each solution and its corresponding niche, so the larger this value, the smaller the density around the solution. Finally, the two values are respectively normalized. The diversity quality is obtained on the grounds of the following formula:

$$D = \frac{1}{\frac{1}{D_x} + \frac{1}{D_f}} \quad (14)$$

where a higher D value represents the higher diversity quality of a solution. In this way, the comprehensive consideration of the distribution of solutions in both two spaces can enhance diversity performance and search capability.

Algorithm 4 Diversity Quality Calculation

Input: Population $\mathbf{P}$, N

Output: D

```

1:  $D_f$  = Crowding Distance Calculation ( $\mathbf{P}$ )
2: for  $i = 1 : N$ 
3:   while Flag = false
4:      $Distance(i)$  = Euclidean Distance Calculation ( $\mathbf{P}$ ,  $\mathbf{P}(i)$ )
5:      $Index$  = Descend Sorting ( $Distance(i)$ )
6:      $\mathbf{P}_{sorted} = \mathbf{P}(Index(1 : N))$ 
7:      $Niche_i = \mathbf{P}_{sorted}(1 : k)$ 
8:     Calculate the mean and standard deviation value of
        $Niche(i)$  on each dimension
9:      $R = \text{Gauss}(\text{mean}(i), \text{std}(i))$ 
10:    if  $\mathbf{P}_{sorted}(k+1) \in R$ 
11:      Flag = True
12:    else
13:       $k = k + 1$ 
14:    end if
15:  end while
16:   $D_x$  = MEDx Calculation( $\mathbf{P}(i)$ ,  $Niche(i)$ )
17: Obtain  $D(i)$  by (14)
18: end for

```

D. Mechanism of HDMODE

This subsection provides a comprehensive analysis of the internal mechanisms employed by HDMODE. Within the dual-population framework, the convergence quality of the first population is enhanced through the implementation of a reproduction and archiving method. This addresses the issue

commonly observed in MMOEAs operating in higher-dimensional decision space, characterized by poor search capabilities and a tendency to converge towards local optima.

The interaction of information between the two populations enables the second population to expand the search range. The second population may gradually increase its population size during the evolutionary process, ensuring the maintenance of population diversity. Concurrently, the diversity of the second population is primarily maintained by the environmental selection mechanism, which favors solutions with high diversity qualities. Furthermore, all solutions are divided into multiple clusters utilizing a neighborhood clustering method. This approach employs the density reachable distance to establish neighborhood range for each dimension of the decision variable. Only when all dimensions of a solution fall within the designated range, it is considered part of the neighborhood.

As shown in Fig. 4, the solutions are divided into three categories by clustering in the decision space. The non-dominated solutions of each category are first selected, namely solutions A, B, C, D, and E. Then, a solution with minimum density is reserved in each cluster, that is, its density in decision space is calculated according to the fitting neighborhood of the nearest solution and the density is estimated by calculating the crowding distance in the objective space. Based on the analysis depicted in Fig. 4, it is evident that there exists a neighborhood comprising three solutions for both entities A and B. However, A exhibits the smallest neighborhood distance among them. Moreover, the calculated average of all the neighborhood distances pertaining to A is significantly lower when compared to that of B. Therefore, it can be concluded from the following equation (15) that the D_x value of A is smaller than the D_x value of B.

$$D_x = \frac{\min(ED(x, x_n)) \cdot \sum_{i=1}^n ED(x, x_n)}{n} \quad (15)$$

where n represents the number of neighborhood solutions and x_n represents the solution in the neighborhood of x . Meanwhile, the crowding distance of A in the objective space is also smaller than that of B. Therefore, it can be calculated from (14) that A has a smaller D value than B, then B has a greater diversity quality than A. This explains why combining the improved environmental selection and the dual-population framework in HDMODE contributes to solving the multimodal multiobjective optimization problems efficiently. In Section V-B, the experimental comparisons are conducted to verify the effectiveness of NMMODE.

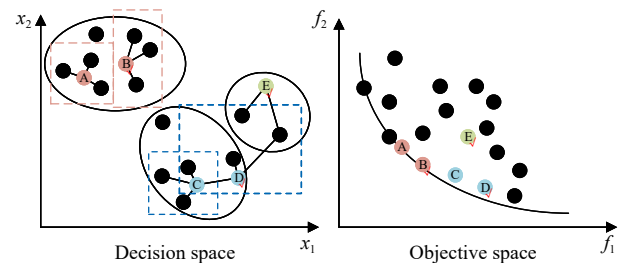


Fig. 4. Illustration of the dual-population framework for solving the higher-dimensional MMOPs.

E. Computational Complexity Analysis

In this section, the computational complexity of HDMODE is outlined. The computational complexity of a single run of HDMODE is primarily determined by the tournament selection mechanism in the reproduction operation (Algorithm 2) and environmental selection mechanism (Algorithm 3). The computational complexity of the tournament selection mechanism is $O(DN^2)$, where D is the number of decision variables and N is the population size. Meanwhile, the computational complexity of Algorithm 3 is mainly determined by non-dominated sorting, neighborhood-based clustering, and diversity quality calculations (Algorithm 4). The computational complexities of non-dominated sorting and neighborhood-based clustering are $O(N\log_{m-2}N)$ [19] and $O(DN^2)$ [50] respectively, while the computational complexity of Algorithm 4 is $O(DN^2)$. In Algorithm 3, the worst-case computational complexity is $O(nlfDN^2)$, where nlf is the number of solutions in the last Pareto front. Theoretically, nlf is at most N , so the worst-case computational complexity of Algorithm 3 is $O(DN^3)$. In summary, the proposed HDMODE has a computational complexity of $O(DN^3)$.

V. EXPERIMENTS AND ANALYSIS

A. Performance Indicators and Experimental Parameter Settings

The experiments are executed on fifteen test function instances termed HDMMF1–HDMMF15. In order to effectively compare the performance of MMOEAs, three performance indicators are used to measure the comprehensive performance, namely inverted generational distance in the objective space (IGD) [66], inverted generational distance in the decision space (IGDX) [67], and hypervolume (HV) [68]. In fact, according to [69], using the IGDX indicator can simultaneously reflect the convergence and diversity performance of MMOEAs in the decision space, while IGD and HV can reflect the convergence and diversity performance in the objective space.

To validate the effectiveness of HDMODE and its internal mechanism, two comparison experiments are conducted in this paper. First, HDMODE is compared with its six variants with different mechanisms. Second, HDMODE compares with other competitive MMOEAs to validate its superiority on difficult MMOPs, such as TriMOEA-TA&R [39], MMOEA/DC [50], MMEAWI [21], MO_Ring_PSO_SCD [37], HREA [40], and DN-NSGA-II [47]. All the parameters of compared MMOEAs are set in light of their original papers. In HDMODE, the penalty neighborhood radius r is set to 0.2, and the number of individuals in the fitting neighborhood k is set to 4. The experimental results of parameter validation can be found in Section S-I of Supplementary Material. To conduct unbiased comparison experiments, the population size is set to 200; the maximum quantity of function evaluations is set to 200000; all the experiments are executed independently 31 times.

B. Comparison With Six Competitive MMOEAs

In this section, six state-of-the-art MMOEAs are systematically compared with HDMODE on fifteen test functions

adopted. The results of Wilcoxon ranksum test with $p < 0.05$ are shown in Tables IV and V, where the mean of IGDX and IGD values acquired by all compared MMOEAs over 31 standalone runs are listed. The HV values obtained by all compared MMOEAs and the comparative results of CPU computation time for these MMOEAs are presented in Section S-III of the Supplementary Material. These algorithms are also tested on HDMMF7 and HDMMF15 with different decision variables dimensions and the IGDX and IGD results are displayed in Figs. 7 and 8.

Tables IV and V present that HDMODE obtains the minimum indicator values in 10 and 7 of 15 instances and it shows superior performance over other competitive MMOEAs in decision space and objective space. From Table IV, it can be seen that MMOEA/DC achieved the smallest results in three functions and demonstrated significant advantages over other comparative algorithms in HDMMF3, 4, 10, 12, 14, and 15. The diversity-first strategy in MMOEA/DC prioritizes preserving the optimal solutions in each cluster and it uses the HAD value to select parents, avoiding premature convergence of the population. However, the diversity-first operation results in insufficient search range and convergence performance, further leading to poor performance in most problems.

In Table V, it can be observed that HDMODE still exhibits significant advantages in the objective space compared to other comparison algorithms. Additionally, DN-NSGA-II, TriMOEA-TA&R, and MMEAWI show better performance compared to other comparison algorithms. MMEAWI enhances solution quality by effectively combining convergence external archives with local convergence quality. In contrast, the performance of TriMOEA-TA&R relies on the accuracy of decision variable analysis, which may lead to unstable performance. For example, it performs better in HDMMF11 but shows noticeably poorer performance in some problems such as HDMMF5 and HDMMF10. On the contrary, DN-NSGA-II has relatively stable performance.

It can be seen from Figs. 5 and 6 that the mean IGD and IGDX values obtained by all compared MMOEAs gradually become larger as the dimension of decision variables increases. It indicates that the performance of these algorithms will gradually deteriorate. However, from the performance changes comparison of MMOEAs in Figs. 5 and 6, it can be observed that as the decision variable dimension increases, the variations in MMOEAs on HDMMF7 are smaller compared to those on HDMMF15. This indicates that the solving difficulty of HDMMF15 is more sensitive to changes in decision variable dimensions. Moreover, HDMODE achieved the best results in almost all the dimensions of decision variables tested. In addition, as shown in Fig. 5, when the dimensions of decision variables in HDMMF15 are 30 and 50, HDMODE achieves similar performance to other compared algorithms in the objective space but it has better performance in the decision space. When the dimensions of decision variables are 75 and 100, the IGD and IGDX values obtained by HDMODE increase smaller than that of other algorithms, which demonstrates it can alleviate the impact of increasing the dimension of decision variables.

TABLE IV
AVERAGE **IGDX** VALUES OVER 31 RUNS ON HDMMFs, IN WHICH MINIMUM INDICATOR FOR EACH HDMMF IS PRESENTED IN BOLD

Problem	DN-NSGA-II [47]	MMOEA/DC [50]	MO_Ring_PSO_SCD [37]	TriMOEA-TA&R [39]	MMEA-WI [21]	HREA [40]	HDMMODE
HDMMF1	1.78e+01 (2.59e+00) +	1.00e+01 (2.37e+00) −	1.39e+01 (1.62e+00) +	1.81e+01 (2.64e+00) +	2.09e+01 (4.01e+00) +	8.04e+00 (6.42e−01) −	1.12e+1 (1.78e+00)
HDMMF2	4.67e+00 (1.70e−01) +	2.30e−01 (3.59e−02) +	2.04e+00 (3.23e−01) +	4.77e+00 (2.52e−01) +	4.87e+00 (2.78e−01) +	1.10e+01 (3.41e−01) +	1.02e−01 (3.32e−02)
HDMMF3	5.84e+00 (1.55e+00) +	2.88e+00 (6.69e−01) +	5.04e+00 (7.64e−01) +	5.71e+00 (1.83e+00) +	7.66e+00 (3.64e−01) +	3.48e+00 (7.97e−01) +	2.13e+00 (2.60e−01)
HDMMF4	7.99e+00 (4.05e+00) +	2.80e+00 (1.29e−01) −	7.91e+00 (3.73e+00) +	8.09e+00 (4.65e+00) +	1.77e+01 (4.15e+00) +	1.89e+01 (2.07e+00) +	3.90e+00 (7.32e−01)
HDMMF5	2.32e+01 (3.74e+00) =	2.49e+01 (1.73e+00) =	3.35e+01 (4.28e+00) +	3.50e+01 (6.44e+00) +	4.17e+01 (7.75e+00) +	3.43e+01 (4.54e+00) +	2.40e+01 (2.57e+00)
HDMMF6	1.49e+01 (3.84e+00) +	1.98e+01 (1.72e+00) +	1.53e+01 (2.85e+00) +	1.53e+01 (4.32e+00) +	2.81e+01 (3.48e+00) +	1.91e+01 (7.32e+00) +	1.35e+01 (3.33e+00)
HDMMF7	1.70e+01 (4.23e+00) +	2.09e+01 (2.36e+00) +	2.40e+01 (2.88e+00) +	1.80e+01 (9.16e+00) +	2.42e+01 (5.21e+00) +	3.38e+01 (1.21e+01) +	1.12e+01 (1.71e+00)
HDMMF8	9.33e+00 (1.04e+00) +	7.39e+00 (4.77e−01) +	6.14e+00 (4.53e−01) +	9.12e+00 (1.17e+00) +	1.01e+01 (1.38e+00) +	1.06e+01 (1.55e+00) +	4.59e+00 (5.45e−01)
HDMMF9	9.70e+00 (3.85e+00) +	3.18e+00 (7.02e−01) +	3.45e+00 (8.59e−01) +	9.76e+00 (2.75e+00) +	6.09e+00 (2.20e+00) +	2.28e+00 (6.24e−02) +	2.09e+00 (3.67e−01)
HDMMF10	9.28e+00 (1.45e+00) +	1.84e+00 (3.10e−01) =	6.32e+00 (9.99e−01) +	8.70e+00 (1.80e+00) +	1.20e+01 (1.86e+00) +	8.69e+00 (2.26e+00) +	3.24e+00 (3.06e+00)
HDMMF11	1.43e+01 (2.01e+00) =	1.48e+01 (1.40e+00) +	1.43e+01 (1.20e+00) =	1.54e+01 (3.46e+00) +	2.19e+01 (2.15e+00) +	2.62e+01 (6.50e−01) +	1.35e+01 (1.25e+00)
HDMMF12	1.51e+01 (1.83e+00) +	1.07e+01 (1.23e+00) =	1.25e+01 (1.10e+00) +	1.51e+01 (2.31e+00) +	2.17e+01 (3.08e+00) +	1.55e+01 (1.03e+00) +	1.12e+01 (1.16e+00)
HDMMF13	2.51e+01 (3.54e+00) +	2.67e+01 (2.98e+00) +	3.21e+01 (5.81e−01) +	2.40e+01 (1.52e+00) +	3.03e+01 (3.50e+00) +	6.01e+01 (1.19e+00) +	1.91e+01 (6.39e−01)
HDMMF14	1.32e+01 (5.45e+00) +	6.37e+00 (7.04e−01) +	1.09e+01 (1.47e+00) +	1.48e+01 (3.97e+00) +	2.08e+01 (2.77e+00) +	2.02e+01 (2.73e+00) +	6.04e+00 (9.65e−01)
HDMMF15	2.11e+01 (2.17e+00) +	1.92e+01 (4.65e+00) +	2.24e+01 (1.50e+00) +	2.17e+01 (3.59e+00) +	2.81e+01 (2.61e+00) +	3.20e+01 (1.63e+00) +	1.31e+01 (1.77e+00)
+/=/−	13/2/0	10/3/2	14/1/0	15/0/0	15/0/0	14/0/1	

TABLE V
AVERAGE **IGD** VALUES OVER 31 RUNS ON HDMMFs, IN WHICH MINIMUM INDICATOR FOR EACH HDMMF IS PRESENTED IN BOLD

Problem	DN-NSGA-II [47]	MMOEA/DC [50]	MO_Ring_PSO_SCD [37]	TriMOEA-TA&R [39]	MMEA-WI [21]	HREA [40]	HDMMODE
HDMMF1	9.47e+00 (3.28e+00) +	6.28e+00 (4.69e+00) =	8.37e+00 (4.92e−01) +	2.95e+00 (1.11e+00) −	9.19e+00 (2.10e+00) +	9.56e+00 (1.03e+00) +	7.81e+00 (1.16e+00)
HDMMF2	1.01e−01 (8.33e−02) =	8.96e−02 (6.72e−03) +	5.19e+02 (1.76e+02) +	1.32e−01 (5.99e−02) +	5.21e−02 (3.52e−02) =	1.62e+04 (1.05e+04) +	6.27e−02 (4.28e−02)
HDMMF3	4.30e−03 (2.34e−04) −	1.94e−02 (3.54e−03) +	7.99e−02 (7.52e−03) +	1.01e−02 (1.29e−03) +	4.24e−03 (6.79e−04) −	8.80e−02 (2.57e−02) +	7.78e−03 (8.52e−04)
HDMMF4	1.21e+01 (1.56e+01) +	3.15e−02 (2.01e−03) +	8.95e+00 (7.72e+00) +	7.78e+00 (1.22e+01) +	2.19e+00 (3.27e+00) +	1.47e+00 (1.04e+00) +	3.25e−02 (6.20e−02)
HDMMF5	9.30e+09 (2.78e+10) +	3.13e+26 (1.58e+26) +	1.09e+07 (2.56e+07) −	1.15e+34 (6.39e+34) +	1.30e+12 (3.29e+12) +	2.39e+11 (1.01e+12) +	4.51e+07 (1.90e+08)
HDMMF6	7.06e+00 (3.00e+01) −	2.30e+05 (4.52e+05) +	2.57e+01 (4.79e+01) +	2.36e+02 (1.27e+03) +	4.14e+01 (6.18e+01) +	9.50e+06 (4.63e+07) +	3.16e+01 (9.26e+01)
HDMMF7	5.20e+00 (2.62e+01) +	1.03e+03 (1.36e+03) +	1.61e+02 (7.09e+01) +	1.05e+01 (2.69e+01) +	2.76e+11 (1.54e+12) +	2.04e+15 (8.33e+15) +	1.22e−01 (1.60e−01)
HDMMF8	1.29e−02 (3.91e−03) −	3.61e+00 (7.38e−01) +	1.89e+00 (3.78e−01) +	1.14e+00 (2.94e+00) +	1.16e−01 (9.57e−02) −	3.17e+00 (9.55e−01) +	1.03e+00 (1.92e−01)
HDMMF9	8.17e−01 (5.82e−01) +	5.19e−01 (1.54e−01) +	6.87e+00 (2.29e+00) +	5.19e−01 (1.44e−01) +	7.19e−01 (3.88e−01) +	1.76e+00 (3.72e−01) +	2.64e−01 (2.20e−01)
HDMMF10	1.53e+02 (1.18e+02) +	1.51e+00 (6.70e−01) +	6.60e+03 (2.55e+03) +	1.76e+02 (1.17e+02) +	2.02e+02 (1.31e+02) +	2.62e+02 (1.99e+02) +	1.17e−02 (3.78e−03)
HDMMF11	4.94e+03 (1.13e+04) +	7.79e+02 (2.06e+02) −	4.56e+02 (1.60e+02) −	2.09e+01 (2.40e+01) −	7.51e+01 (1.03e+02) −	5.73e+03 (1.61e+03) +	1.96e+03 (5.50e+02)
HDMMF12	8.96e+01 (2.59e+01) +	1.04e+02 (1.76e+01) +	2.72e+02 (5.82e+01) +	8.78e+01 (3.76e+01) +	1.46e+02 (9.64e+01) +	5.44e+02 (2.76e+02) +	5.16e+01 (1.40e+01)
HDMMF13	1.73e+04 (1.02e+04) +	2.04e+05 (1.08e+05) +	3.19e+06 (2.32e+05) +	1.94e+04 (1.19e+04) −	1.47e+04 (1.01e+04) −	4.98e+07 (9.75e+06) +	2.22e+04 (1.04e+04)
HDMMF14	2.15e−01 (1.54e−01) +	1.97e−02 (8.94e−03) +	4.14e+02 (9.75e+01) +	4.02e−01 (2.90e−01) +	2.95e+01 (2.29e+01) +	4.55e+03 (1.11e+03) +	5.41e−03 (3.80e−04)
HDMMF15	8.81e+00 (1.23e+00) +	1.16e+01 (6.94e−01) +	1.18e+01 (6.75e−01) +	7.93e+00 (2.89e+00) −	9.87e+00 (1.36e+00) +	1.80e+01 (2.99e+00) +	7.95e+00 (2.12e−01)
+/=/−	11/1/3	13/1/1	13/0/2	11/0/4	10/1/4	15/0/0	

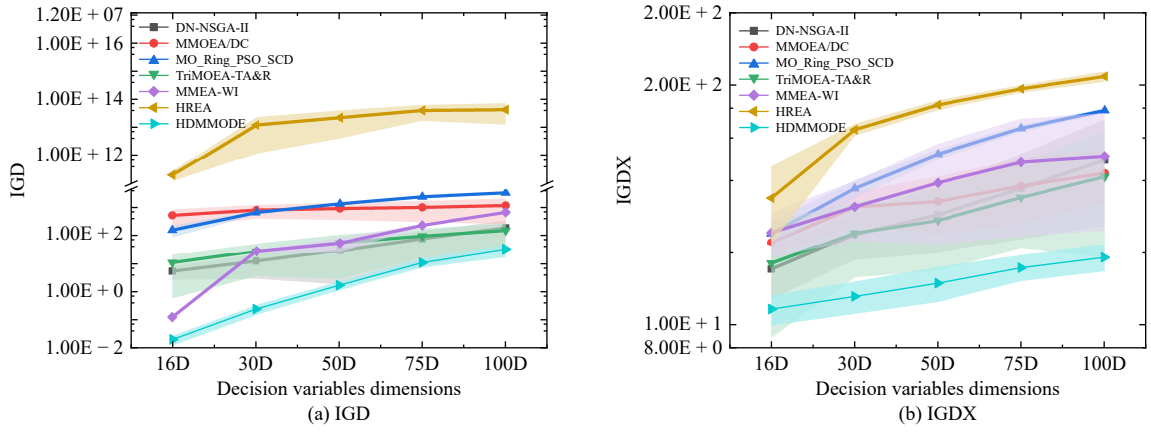


Fig. 5. Presentation of IGD and IGDX values acquired by six competitive MMOEAs on HDMMF7 with 16D, 30D, 50D, 75D, and 100D.

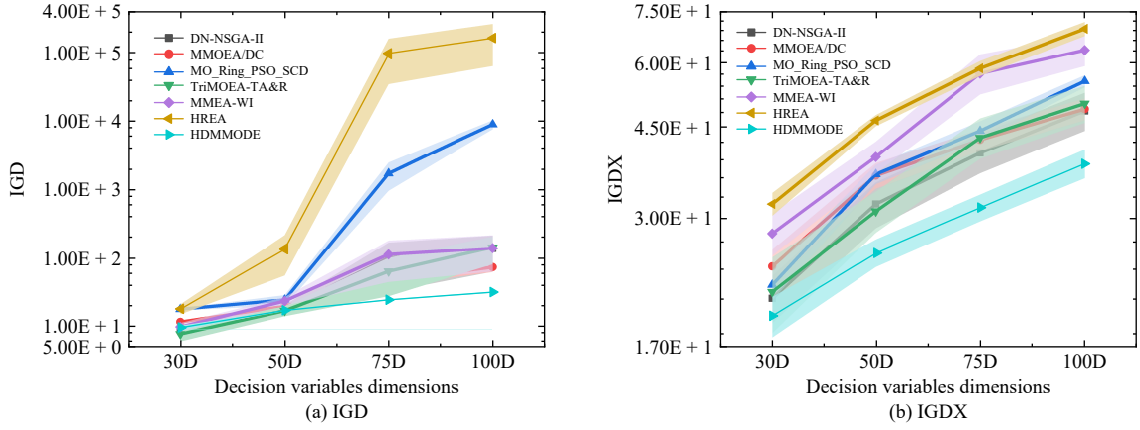


Fig. 6. Comparison of IGD and IGDX results acquired by six competitive MMOEAs on HDMMF15 with 30D, 50D, 75D, and 100D.

The reasons why the compared MMOEAs perform worse than HDMMODE on the MMOPs adopted are given as follows. DN-NSGA-II introduces a niching method to generate new solutions and adopts the nondominated sorting method to select solutions with better convergence, which brings limited diversity performance in decision space to remove it from local optimal regions. Although MO_Ring_PSO_SCD considers the crowding distance of solutions in the decision space, it may be inaccurate as the dimension increases, leading to the elimination of distant and potential solutions. Furthermore, due to its restricted population information exchange, this further leads to a relatively slow convergence speed of the population, making it difficult to obtain high-quality non-dominated solutions. MMOEA/DC prefers to maintain solutions with good diversity and it aims to preserve both global and local PSs, which leads to convergence performance in the objective and decision space being insufficient to find and maintain multiple Pareto optimal solutions in difficult MMOPs, such as HDMMF5, HDMMF6, and HDMMF7. MMEA-WI and HREA are easily trapped in the local optimal regions, which weakens its searchability in the decision space and significantly deteriorates its performance on several test functions, such as HDMMF5 and HDMMF7. TriMOEA-TA&R still prefers the convergence quality of solutions in the objective space and it cannot effectively solve the MMOPs with nonseparable decision variables. Because it is highly likely that the population will not converge to the true PF in

the objective space, if the decision variables are grouped incorrectly. This also causes it to fail to retain multiple Pareto optimal solutions in the decision space. Therefore, it can be reasonably concluded from Tables IV and V, Figs. 5 and 6 that HDMMODE has greater advantages than the compared MMOEAs for solving the higher-dimensional MMOPs.

C. Experimental Validation of the Effectiveness of HDMMODE

Six variants of HDMMODE are employed for comparative experiments in this section, such as variants with other environmental selection mechanisms and reproduction mechanisms, variants without the dual-population framework, and so on. These variants are tested on fifteen HDMMFs and their IGDX indicator are calculated by executing a Wilcoxon ranksum test with $p < 0.05$, which is shown in Table VI. The symbols +, =, and - mean that the performance indicator of the proposed algorithm is significantly better than, similar, and worse than other comparison algorithms, respectively. The IGD values obtained by these algorithms are displayed in Tables S-II and S-III in the Supplementary Material. Table VI shows the detailed IGDX values acquired by the proposed algorithm and its six variants. Detailed explanations of all variants of HDMMODE are shown in Table VII.

From Table VI, it can be concluded that HDMMODE achieves the best performance in 8 of 15 test functions and it performs significantly better than other variants. In Table VI, **Variant1** adopts a tournament selection strategy to select par-

TABLE VI
AVERAGE **IGDX** VALUES OVER 31 RUNS ON HDMMFs, IN WHICH MINIMUM INDICATOR FOR EACH HDMMF IS PRESENTED IN BOLD

Problem	Variant1	Variant2	Variant3	Variant4	Variant5	Variant6	HDMMODE
HDMMF1	1.89e+01 (2.61e+00) +	1.42e+01 (2.03e+00) +	2.02e+01 (2.27e+00) +	1.38e+01 (6.51e+00) =	1.28e+01 (1.53e+00) +	2.09e+01 (2.67e+00) +	1.12e+1 (1.78e+00)
HDMMF2	4.67e+00 (2.33e-01) +	1.81e+00 (1.69e+00) +	4.79e+00 (2.35e-01) +	6.12e-01 (1.50e+00) +	3.53e-01 (1.32e-01) +	4.92e+00 (2.80e-01) +	1.02e-01 (3.32e-02)
HDMMF3	5.98e+00 (1.50e+00) +	3.31e+00 (8.46e-01) +	6.65e+00 (1.25e+00) +	2.21e+00 (2.02e-01) =	2.45e+00 (4.36e-01) +	8.73e+00 (1.11e+00) +	2.13e+00 (2.60e-01)
HDMMF4	6.89e+00 (4.11e+00) +	4.22e+00 (1.95e+00) =	1.48e+01 (4.29e+00) +	3.40e+00 (1.06e-02) -	4.52e+00 (1.85e+00) +	1.42e+01 (4.13e+00) +	3.90e+00 (7.32e-01)
HDMMF5	2.38e+01 (4.21e+00) =	2.83e+01 (5.30e+00) +	2.93e+01 (1.78e+00) +	2.43e+01 (2.37e+00) =	2.53e+01 (3.24e+00) =	2.92e+01 (4.07e+00) +	2.40e+01 (2.57e+00)
HDMMF6	1.20e+01 (2.56e+00) -	1.19e+01 (2.65e+00) -	1.60e+01 (2.71e+00) +	1.50e+01 (2.90e+00) +	1.20e+01 (2.11e+00) -	1.62e+01 (2.46e+00) +	1.35e+01 (3.33e+00)
HDMMF7	1.80e+01 (4.81e+00) +	1.75e+01 (5.28e+00) +	1.60e+01 (4.47e+00) +	1.23e+01 (1.30e+00) +	1.38e+01 (2.80e+00) +	1.55e+01 (3.89e+00) +	1.12e+01 (1.71e+00)
HDMMF8	9.26e+00 (7.70e-01) +	7.81e+00 (1.06e+00) +	5.20e+00 (5.25e-01) +	4.89e+00 (5.44e-01) +	5.01e+00 (7.10e-01) +	5.17e+00 (5.37e-01) +	4.59e+00 (5.45e-01)
HDMMF9	8.36e+00 (3.91e+00) +	9.11e+00 (3.68e+00) +	4.56e+00 (1.23e+00) +	3.46e+00 (8.95e-01) +	4.47e+00 (1.24e+00) +	4.39e+00 (1.41e+00) =	2.09e+00 (3.67e-01)
HDMMF10	8.80e+00 (1.39e+00) +	7.26e+00 (1.75e+00) +	1.05e+01 (2.08e+00) +	5.02e+00 (2.28e+00) +	3.27e+00 (1.88e+00) =	9.91e+00 (2.18e+00) +	3.24e+00 (3.06e+00)
HDMMF11	1.65e+01 (2.03e+00) +	1.34e+01 (2.64e+00) =	1.29e+01 (8.45e-01) -	1.42e+01 (1.49e+00) +	1.28e+01 (1.90e+00) -	2.27e+01 (1.38e+00) +	1.35e+01 (1.25e+00)
HDMMF12	1.41e+01 (2.13e+00) +	1.11e+01 (1.42e+00) =	1.60e+01 (2.34e+00) +	1.20e+01 (1.33e+00) +	1.17e+01 (1.37e+00) =	1.72e+01 (3.43e+00) +	1.12e+01 (1.16e+00)
HDMMF13	2.61e+01 (3.32e+00) +	2.24e+01 (1.22e+00) +	2.95e+01 (2.45e+00) +	1.86e+01 (1.48e+00) -	2.01e+01 (1.35e+00) +	2.94e+01 (2.91e+00) +	1.91e+01 (6.39e-01)
HDMMF14	1.51e+01 (4.31e+00) +	1.12e+01 (3.54e+00) +	1.39e+01 (4.33e+00) +	5.79e+00 (5.96e-01) -	6.68e+00 (9.94e-01) +	1.53e+01 (3.59e+00) +	6.04e+00 (9.65e-01)
HDMMF15	2.23e+01 (2.86e+00) +	2.10e+01 (2.08e+00) +	2.57e+01 (2.59e+00) +	2.00e+01 (2.74e+00) +	2.11e+01 (2.71e+00) +	2.46e+01 (2.73e+00) +	1.31e+01 (1.77e+00)
+/-/-	13/1/1	11/3/1	14/0/1	9/3/3	10/3/2	14/1/0	

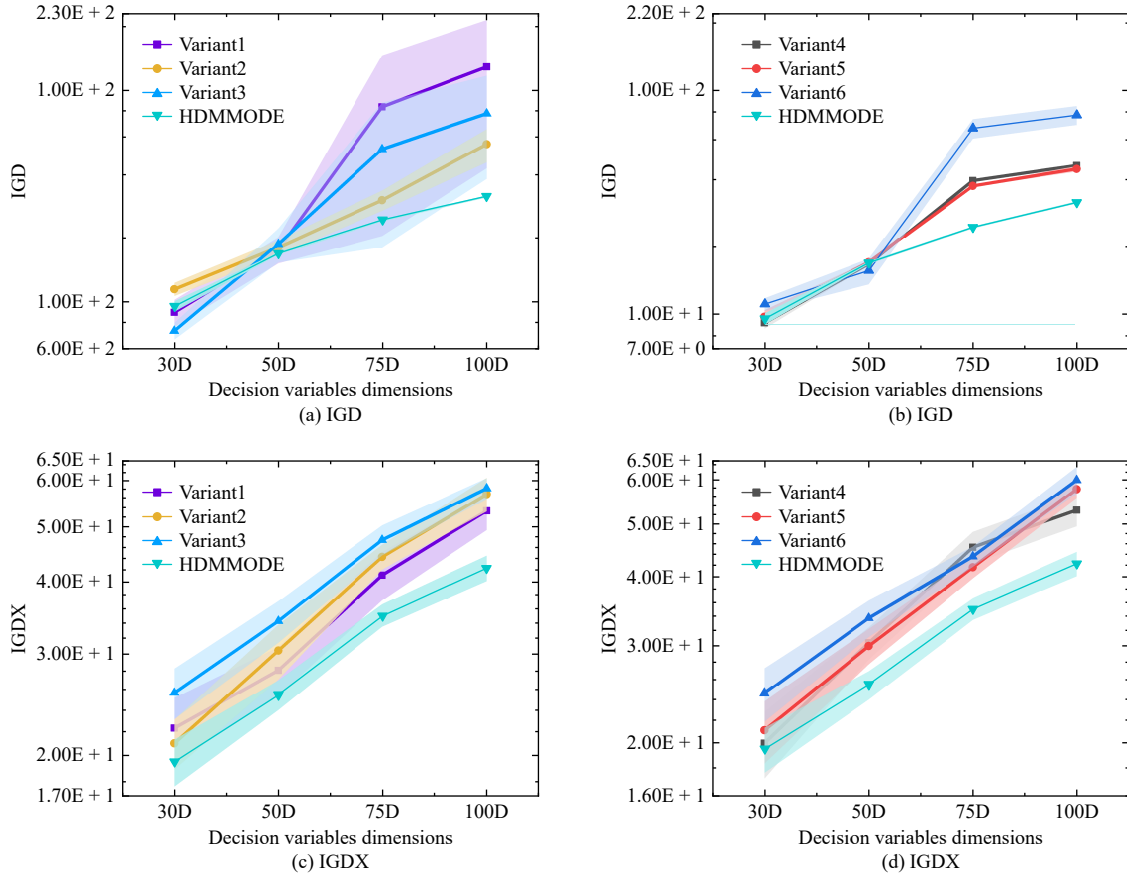


Fig. 7. Comparison of IGD and IGDX values obtained by the six variants on HDMMF15 with 30D, 50D, 75D, and 100D.

TABLE VII
DETAILED EXPLANATIONS OF ALL VARIANTS OF HDMMODE

Algorithm	Explanation
Variant1	It adopts the niche-based tournament selection strategy in [47]
Variant2	It uses DE/current-to-rand/1 and DE/current-to-best/1/bin
Variant3	It updates SCD-based environmental selection mechanism in [37]
Variant4	It uses the single population framework to evolve
Variant5	It employs the original dual-population framework
Variant6	It updates the population without density estimation

ents and utilizes genetic algorithm (GA) operator [70] to generate offspring. In contrast, **Variant2** uses two DE strategies to generate offspring. From Table VI, it can be observed that **Variant2** has smaller IGDX values than **Variant1** in 13 out of 15 functions. The population distribution in Fig. 8 also reveals that **Variant2** retains a true PS, while **Variant1** does not. This indicates that using DE operator to generate offspring contributes to improving the search capabilities of the population. According to the IGDX results, **Variant3** performs worse than **Variant2** in 12 functions. This is because its environmental selection mechanism still sequentially selects solutions based on their Pareto ranks, maintaining a convergence-first selection. This leads to the deletion of potentially dominating solutions, making it challenging for the population to expand to other effective search directions. In comparison, **Variant4** and **Variant5** achieve three and two best results, respectively, out of 15 problems. In addition, the performance

of these two variants surpasses **Variant1**, **Variant2**, and **Variant3**. This suggests that the proposed reproduction strategy and improved environmental selection mechanism effectively enhance the search capabilities.

D. Discussion on HDMMODE

To analyze the differences in the internal mechanisms of HDMMODE more directly, this paper extends HDMMF15 to four dimensions: 30, 50, 75, and 100, to test the performance variations of the algorithm, with the results presented in Fig. 7. Meanwhile, to more intuitively show the differences in equivalent PSs obtained by these compared algorithms, the t-SNE algorithm [71] is employed to reduce the dimensions of decision variables and visualize the true PSs and the PSs acquired by compared variants on HDMMF2. Fig. 8 shows these results. The red dots represent the true PSs and the blue dots represent the PSs obtained by the algorithms.

HDMMODE also has significant advantages over **Variant4**, which proves that adopting two populations to explore the remaining search space respectively is more helpful for guaranteeing the search capability. The reason why HDMMODE performs better than **Variant5** is that the second population will also obtain more solutions to search the remaining decision space as the number of solutions in the first population reduces, which may improve the diversity of the whole population. The comparison results with HDMMODE validating the effectiveness of the dual-population framework and neighborhood penalty strategy proposed in this paper. However,

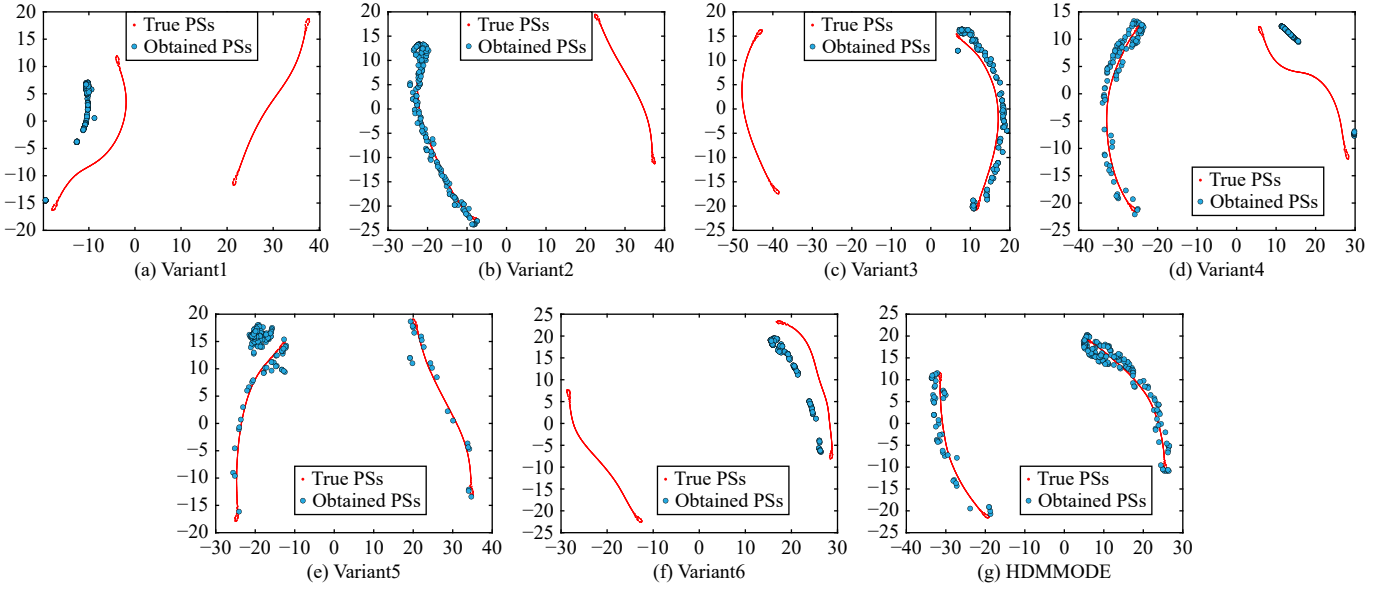


Fig. 8. The visualization of the final population obtained by the six variants and true PSs on HDMMF2, which is obtained by reducing the dimension of decision variables through t-SNE algorithm. The red dots denotes the reference solution on the true PSs and the blue dots denotes the solutions obtained by different algorithms.

Variant6, due to not considering the diversity quality of solutions in the decision space, retains only solutions with good convergence quality during population updates, resulting in a lack of diversity and the population getting trapped in a local optimal region. Figs. 8(f), 7(b), and 7(d) also illustrate that insufficient diversity leads to a severe degradation in performance after expanding the decision space.

As can be seen from Fig. 7(a), the IGD values of these variants are similar, when the dimension of decision variables is lower than 50. Then the performance of **Variant1** and **Variant3** deteriorate significantly as the dimensions increase, while the performances of HDMMODE and the remaining two variants do not. This indicates that using the DE operator to generate offspring helps expand the search range of the population, thereby more effectively preventing the population from falling into local optima. In addition, it can be seen in Fig. 7(b) that HDMMODE achieves better performance compared with other variants in the decision space with different dimensions. Figs. 7(b) and 7(d) show that the standard deviations of IGDX and IGD values of **Variant4**, **Variant5**, and **Variant6** are all relatively small, which indicates that they have better stability and convergence performance. Meanwhile, three variants have enough diversity to mitigate the deterioration of performance caused by the increasing dimensionality of the decision variables.

It can be seen intuitively from Fig. 8 that **Variant1** and **Variant6** do not find the true PSs, while **Variant2**, **Variant3** find only one true PS. **Variant4** is well-distributed on one PS and retains some solutions near another PS. On the other hand, **Variant5** has solutions on two true PSs, but the main population clusters in an area near one PS. The comparison between **Variant4** and **Variant1** and **Variant2** suggests that the offspring generation and environmental selection strategies proposed in this paper can help the population achieve better convergence and diversity. From Figs. 7 and 8, it is proven that

HDMMODE has advantages over them in solving higher-dimensional MMOPs.

VI. CONCLUSION

This paper focuses on studying multimodal multiobjective optimization problems with higher-dimensional decision variables and designing specific evolutionary algorithms to solve them. A novel MMO benchmark test set is developed, incorporating diverse characteristics such as higher-dimensional decision variables, intricate fitness landscapes, controllable differences in dimensions among different PSs, and intricate coupling relationships among decision variables. The aim is to more accurately simulate the features of real-world problems. Consequently, this test problem set introduces heightened convergence challenges and more intricate fitness landscapes, thereby presenting increased difficulty for the diversity and convergence performance of MMOEAs in both decision and objective spaces. Furthermore, the proposed benchmark test framework encompasses adaptable component functions, providing additional support for researchers to delve into the characteristics of high-dimensional MMOPs and devising more effective algorithms. In this paper, we introduce a novel multimodal multiobjective optimization evolutionary algorithm, named HDMMODE, utilizing a dual-population framework and an enhanced environmental selection mechanism, with the aim of addressing this class of problem preliminarily.

In the proposed algorithm, the two populations work collaboratively. The first population aims to converge towards the true Pareto front by utilizing an external convergence archive and a population reduction mechanism. Meanwhile, the second population focuses on exploring the remaining search space to maintaining multiple equivalent Pareto optimal solutions in the decision space. Furthermore, the improved environmental selection mechanism incorporates a diversity quality assessment method that accurately identifies and retains potential and distant solutions in the decision space. This

enhancement improves the search ability and mitigates performance deterioration as the dimensionality of the decision variables increases. To evaluate the effectiveness of HDMODE, we conduct experimental comparisons with six variants of the algorithm using adopted test functions. Additionally, we perform comparative experiments with six competitive multimodal multiobjective evolutionary algorithms. The results consistently demonstrate that HDMODE outperforms both its variants and the competitive algorithms in terms of decision and objective space performance.

In future work, we will further investigate the performance of evolutionary algorithms on more complex high-dimensional multimodal multi-objective optimization problems. We will design a subpopulation generation strategy with adaptive step sizes to effectively balance the global search capability and local explore capability of MMOEAs, further improving the quality of generated solutions. Additionally, the proposed algorithm will be extended to address more complex optimization problems.

REFERENCES

- [1] K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan, "A fast and elitist multiobjective genetic algorithm: NSGA-II," *IEEE Trans. Evol. Comput.*, vol. 6, no. 2, pp. 182–197, Apr. 2002.
- [2] F. Kudo, T. Yoshikawa, and T. Furuhashi, "A study on analysis of design variables in Pareto solutions for conceptual design optimization problem of hybrid rocket engine," in *Proc. IEEE Congr. Evolutionary Computation*, New Orleans, USA, 2011, pp. 2558–2562.
- [3] D.-H. Cho, H.-K. Jung, and C.-G. Lee, "Induction motor design for electric vehicle using a niching genetic algorithm," *IEEE Trans. Ind. Appl.*, vol. 37, no. 4, pp. 994–999, Jul.–Aug. 2001.
- [4] X. Yao, W. Li, X. Pan, and R. Wang, "Multimodal multi-objective evolutionary algorithm for multiple path planning," *Comput. Ind. Eng.*, vol. 169, p. 108145, Jul. 2022.
- [5] J. Liang, C. Yue, G. Li, B. Qu, P. N. Suganthan, and K. Yu, "Problem definitions and evaluation criteria for the CEC 2021 on multimodal multiobjective path planning optimization," Computational Intelligence Laboratory, Zhengzhou University, Zhengzhou, China, Computational Intelligence Laboratory, Nanyang Technological University, Singapore, Tech. Rep. 202012, 2020.
- [6] C. T. Yue, J. J. Liang, B. Y. Qu, K. J. Yu, and H. Song, "Multimodal multiobjective optimization in feature selection," in *Proc. IEEE Congr. Evolutionary Computation*, Wellington, New Zealand, 2019, pp. 302–309.
- [7] K. Jha and S. Saha, "Incorporation of multimodal multiobjective optimization in designing a filter based feature selection technique," *Appl. Soft Comput.*, vol. 98, p. 106823, Jan. 2021.
- [8] M. Wang, X. Li, and L. Chen, "An enhance multimodal multiobjective optimization genetic algorithm with special crowding distance for pulmonary hypertension feature selection," *Comput. Biol. Med.*, vol. 146, p. 105536, Jul. 2022.
- [9] E. Pérez, M. Posada, and A. Lorenzana, "Taking advantage of solving the resource constrained multi-project scheduling problems using multimodal genetic algorithms," *Soft Comput.*, vol. 20, no. 5, pp. 1879–1896, May 2016.
- [10] T. Yang, S.-Y. Lin, Y.-H. Hung, and C.-C. Hong, "A study on the optimization of in-process inspection procedure for active pharmaceutical ingredients manufacturing process," *Sustainability*, vol. 14, no. 6, p. 3706, Mar. 2022.
- [11] K.-L. Zhao, J. Liu, X.-G. Zhou, J.-Z. Su, Y. Zhang, and G.-J. Zhang, "MMpred: A distance-assisted multimodal conformation sampling for *de novo* protein structure prediction," *Bioinformatics*, vol. 37, no. 23, pp. 4350–4356, Dec. 2021.
- [12] D. Pršić, N. Nedić, and V. Stojanović, "A nature inspired optimal control of pneumatic-driven parallel robot platform," *Proc. Inst. Mech. Eng. Part C: J. Mech. Eng. Sci.*, vol. 231, no. 1, pp. 59–71, Jan. 2017.
- [13] V. Stojanovic and N. Nedic, "A nature inspired parameter tuning approach to cascade control for hydraulically driven parallel robot platform," *J. Optim. Theory Appl.*, vol. 168, no. 1, pp. 332–347, Jan. 2016.
- [14] N. Nedic, D. Prsic, L. Dubonjic, V. Stojanovic, and V. Djordjevic, "Optimal cascade hydraulic control for a parallel robot platform by PSO," *Int. J. Adv. Manuf. Technol.*, vol. 72, pp. 5–8, May 1085.
- [15] J. Kennedy and R. Eberhart, "Particle swarm optimization," in *Proc. Int. Conf. Neural Networks*, Perth, Australia, 1995, pp. 1942–1948.
- [16] K. Deb and S. Tiwari, "Omni-optimizer: A procedure for single and multi-objective optimization," in *Proc. 3rd Int. Conf. Evolutionary Multi-Criterion Optimization*, Guanajuato, Mexico, 2005, pp. 47–61.
- [17] K. Qiao, K. Yu, B. Qu, J. Liang, H. Song, and C. Yue, "An evolutionary multitasking optimization framework for constrained multiobjective optimization problems," *IEEE Trans. Evol. Comput.*, vol. 26, no. 2, pp. 263–277, Apr. 2022.
- [18] C. Yue, P. N. Suganthan, J. Liang, B. Qu, K. Yu, Y. Zhu, and L. Yan, "Differential evolution using improved crowding distance for multimodal multiobjective optimization," *Swarm Evol. Comput.*, vol. 62, p. 100849, Apr. 2021.
- [19] T. Zhou, Z. Hu, Q. Su, and W. Xiong, "A clustering differential evolution algorithm with neighborhood-based dual mutation operator for multimodal multiobjective optimization," *Expert Syst. Appl.*, vol. 216, p. 119438, Apr. 2023.
- [20] Y. Liu, H. Ishibuchi, G. G. Yen, Y. Nojima, and N. Masuyama, "Handling imbalance between convergence and diversity in the decision space in evolutionary multimodal multiobjective optimization," *IEEE Trans. Evol. Comput.*, vol. 24, no. 3, pp. 551–565, Jun. 2020.
- [21] W. Li, T. Zhang, R. Wang, and H. Ishibuchi, "Weighted indicator-based evolutionary algorithm for multimodal multiobjective optimization," *IEEE Trans. Evol. Comput.*, vol. 25, no. 6, pp. 1064–1078, Dec. 2021.
- [22] R. Tanabe and H. Ishibuchi, "A decomposition-based evolutionary algorithm for multi-modal multi-objective optimization," in *Proc. 15th Int. Conf. Parallel Problem Solving from Nature*, Coimbra, Portugal, 2018, pp. 249–261.
- [23] K. Qiao, J. Liang, K. Yu, C. Yue, H. Lin, D. Zhang, and B. Qu, "Evolutionary constrained multiobjective optimization: Scalable high-dimensional constraint benchmarks and algorithm," *IEEE Trans. Evol. Comput.*, 2023.
- [24] B. Cao, J. Zhao, Y. Gu, Y. Ling, and X. Ma, "Applying graph-based differential grouping for multiobjective large-scale optimization," *Swarm Evol. Comput.*, vol. 53, p. 100626, Mar. 2020.
- [25] L. M. Antonio, C. A. C. Coello, S. G. Brambila, J. F. González, and G. C. Tapia, "Operational decomposition for large scale multi-objective optimization problems," in *Proc. Genetic and Evolutionary Computation Conf. Companion*, Prague, Czech Republic, 2019, pp. 225–226.
- [26] X. Zhang, Y. Tian, R. Cheng, and Y. Jin, "A decision variable clustering-based evolutionary algorithm for large-scale many-objective optimization," *IEEE Trans. Evol. Comput.*, vol. 22, no. 1, pp. 97–112, Feb. 2018.
- [27] S. Qin, C. Sun, Y. Jin, Y. Tan, and J. Fieldsend, "Large-scale evolutionary multiobjective optimization assisted by directed sampling," *IEEE Trans. Evol. Comput.*, vol. 25, no. 4, pp. 724–738, Aug. 2021.
- [28] Y. Tian, C. Lu, X. Zhang, F. Cheng, and Y. Jin, "A pattern mining-based evolutionary algorithm for large-scale sparse multiobjective optimization problems," *IEEE Trans. Cybern.*, vol. 52, no. 7, pp. 6784–6797, Jul. 2022.
- [29] Y. Tian, C. Lu, X. Zhang, K. C. Tan, and Y. Jin, "Solving large-scale multiobjective optimization problems with sparse optimal solutions via unsupervised neural networks," *IEEE Trans. Cybern.*, vol. 51, no. 6, pp. 3115–3128, Jun. 2021.
- [30] Y. Tian, X. Zheng, X. Zhang, and Y. Jin, "Efficient large-scale multiobjective optimization based on a competitive swarm optimizer," *IEEE Trans. Cybern.*, vol. 50, no. 8, pp. 3696–3708, Aug. 2020.
- [31] Y. Zhang, G.-G. Wang, K. Li, W.-C. Yeh, M. Jian, and J. Dong, "Enhancing MOEA/D with information feedback models for large-scale many-objective optimization," *Inf. Sci.*, vol. 522, pp. 1–16, Jun. 2020.
- [32] H. Chen, R. Cheng, J. Wen, H. Li, and J. Weng, "Solving large-scale many-objective optimization problems by covariance matrix adaptation

- evolution strategy with scalable small subpopulations," *Inf. Sci.*, vol. 509, pp. 457–469, Jan. 2020.
- [33] M. Preuss, B. Naujoks, and G. Rudolph, "Pareto set and EMOA behavior for simple multimodal multiobjective functions," in *Proc. 9th Int. Conf. Parallel Problem Solving from Nature*, Reykjavik, Iceland, 2006, pp. 513–522.
- [34] G. Rudolph, B. Naujoks, and M. Preuss, "Capabilities of EMOA to detect and preserve equivalent Pareto subsets," in *Proc. 4th Int. Conf. Evolutionary Multi-Criterion Optimization*, Matsushima, Japan, 2007, pp. 36–50.
- [35] K. Deb and S. Tiwari, "Omni-optimizer: A generic evolutionary algorithm for single and multi-objective optimization," *Eur. J. Oper. Res.*, vol. 185, no. 3, pp. 1062–1087, Mar. 2008.
- [36] H. Ishibuchi, N. Akedo, and Y. Nojima, "A many-objective test problem for visually examining diversity maintenance behavior in a decision space," in *Proc. 13th Annu. Conf. Genetic and Evolutionary Computation*, Dublin, Ireland, 2011, pp. 649–656.
- [37] C. Yue, B. Qu, and J. Liang, "A multiobjective particle swarm optimizer using ring topology for solving multimodal multiobjective problems," *IEEE Trans. Evol. Comput.*, vol. 22, no. 5, pp. 805–817, Oct. 2018.
- [38] C. Yue, B. Qu, K. Yu, J. Liang, and X. Li, "A novel scalable test problem suite for multimodal multiobjective optimization," *Swarm Evol. Comput.*, vol. 48, pp. 62–71, Aug. 2019.
- [39] Y. Liu, G. G. Yen, and D. Gong, "A multimodal multiobjective evolutionary algorithm using two-archive and recombination strategies," *IEEE Trans. Evol. Comput.*, vol. 23, no. 4, pp. 660–674, Aug. 2019.
- [40] W. Li, X. Yao, T. Zhang, R. Wang, and L. Wang, "Hierarchy ranking method for multimodal multiobjective optimization with local Pareto fronts," *IEEE Trans. Evol. Comput.*, vol. 27, no. 1, pp. 98–110, Feb. 2023.
- [41] Y. Tian, R. Liu, X. Zhang, H. Ma, K. C. Tan, and Y. Jin, "A multipopulation evolutionary algorithm for solving large-scale multimodal multiobjective optimization problems," *IEEE Trans. Evol. Comput.*, vol. 25, no. 3, pp. 405–418, Jun. 2021.
- [42] Y. Liu, H. Ishibuchi, Y. Nojima, N. Masuyama, and K. Shang, "A double-niched evolutionary algorithm and its behavior on polygon-based problems," in *Proc. 15th Int. Conf. Parallel Problem Solving from Nature*, Coimbra, Portugal, 2018, pp. 262–273.
- [43] X. Li, A. Engelbrecht, and M. G. Epitropakis, "Benchmark functions for CEC'2013 special session and competition on niching methods for multimodal function optimization," Evolutionary Computation and Machine Learning Group, RMIT University, Melbourne, Australia, Tech. Rep., 2013.
- [44] B. Y. Qu, J. J. Liang, Z. Y. Wang, Q. Chen, and P. N. Suganthan, "Novel benchmark functions for continuous multimodal optimization with comparative results," *Swarm Evol. Comput.*, vol. 26, pp. 23–34, Feb. 2016.
- [45] Q. Zhang, A. Zhou, S. Zhao, P. N. Suganthan, W. Liu, and S. Tiwari, "Multiobjective optimization test instances for the CEC 2009 special session and competition," University of Essex, Colchester, UK, Nanyang Technological University, Singapore, Tech. Rep. CES-487, 2009.
- [46] K. Deb, L. Thiele, M. Laumanns, and E. Zitzler, "Scalable test problems for evolutionary multiobjective optimization," in *Evolutionary Multiobjective Optimization: Theoretical Advances and Applications*, A. Abraham, L. Jain, and R. Goldberg, Eds. London, UK: Springer, 2005, pp. 105–145.
- [47] J. J. Liang, C. Yue, and B. Y. Qu, "Multimodal multi-objective optimization: A preliminary study," in *Proc. IEEE Congr. Evolutionary Computation*, Vancouver, Canada, 2016, pp. 2454–2461.
- [48] B. Qu, C. Li, J. Liang, L. Yan, K. Yu, and Y. Zhu, "A self-organized speciation based multi-objective particle swarm optimizer for multimodal multi-objective problems," *Appl. Soft Comput.*, vol. 86, p. 105886, Jan. 2020.
- [49] Q. Zhang and H. Li, "MOEA/D: A multiobjective evolutionary algorithm based on decomposition," *IEEE Trans. Evol. Comput.*, vol. 11, no. 6, pp. 712–731, Dec. 2007.
- [50] Q. Lin, W. Lin, Z. Zhu, M. Gong, J. Li, and C. A. C. Coello, "Multimodal multiobjective evolutionary optimization with dual clustering in decision and objective spaces," *IEEE Trans. Evol. Comput.*, vol. 25, no. 1, pp. 130–144, Feb. 2021.
- [51] J. Liang, K. Qiao, C. Yue, K. Yu, B. Qu, R. Xu, Z. Li, and Y. Hu, "A clustering-based differential evolution algorithm for solving multimodal multi-objective optimization problems," *Swarm Evol. Comput.*, vol. 60, p. 100788, Feb. 2021.
- [52] Z. Ding, L. Cao, L. Chen, D. Sun, X. Zhang, and Z. Tao, "Large-scale multimodal multiobjective evolutionary optimization based on hybrid hierarchical clustering," *Knowl.-Based Syst.*, vol. 266, p. 110398, Apr. 2023.
- [53] Y. Peng and H. Ishibuchi, "A decomposition-based large-scale multimodal multi-objective optimization algorithm," in *Proc. IEEE Congr. Evolutionary Computation*, Glasgow, UK, 2020, pp. 1–8.
- [54] J. Liang, H. Lin, C. Yue, K. Yu, Y. Guo, and K. Qiao, "Multiobjective differential evolution with speciation for constrained multimodal multiobjective optimization," *IEEE Trans. Evol. Comput.*, vol. 27, no. 4, pp. 1115–1129, Aug. 2023.
- [55] K. Deb, "Multi-objective genetic algorithms: Problem difficulties and construction of test problems," *Evol. Comput.*, vol. 7, no. 3, pp. 205–230, Sep. 1999.
- [56] Y. Zhou, Y. Xiang, and X. He, "Constrained multiobjective optimization: Test problem construction and performance evaluations," *IEEE Trans. Evol. Comput.*, vol. 25, no. 1, pp. 172–186, Feb. 2021.
- [57] J. J. Liang, B. Y. Qu, and P. N. Suganthan, "Problem definitions and evaluation criteria for the CEC 2014 special session and competition on single objective real-parameter numerical optimization," Computational Intelligence Laboratory, Zhengzhou University, Zhengzhou, China, Computational Intelligence Laboratory, Nanyang Technological University, Singapore, Tech. Rep. 201311, 2013.
- [58] B. Y. Qu, J. J. Liang, P. N. Suganthan, and Q. Chen, "Problem definitions and evaluation criteria for the CEC 2015 competition on single objective multi-niche optimization," Computational Intelligence Laboratory, Zhengzhou University, Zhengzhou, China, Computational Intelligence Laboratory, Nanyang Technological University, Singapore, Tech. Rep. 201411B, 2014.
- [59] J. J. Liang, T. P. Runarsson, E. Mezura-Montes, M. Clerc, P. N. Suganthan, C. A. C. Coello, and K. Deb, "Problem definitions and evaluation criteria for the CEC 2006 special session on constrained real-parameter optimization," Nanyang Technological University, Singapore, Tech. Rep., 2006.
- [60] R. Cheng, Y. Jin, M. Olhofer, and B. Sendhoff, "Test problems for large-scale multiobjective and many-objective optimization," *IEEE Trans. Cybern.*, vol. 47, no. 12, pp. 4108–4121, Dec. 2017.
- [61] R. Storn and K. Price, "Differential evolution—a simple and efficient heuristic for global optimization over continuous spaces," *J. Glob. Optim.*, vol. 11, no. 4, pp. 341–359, Dec. 1997.
- [62] H. Ma, H. Wei, Y. Tian, R. Cheng, and X. Zhang, "A multi-stage evolutionary algorithm for multi-objective optimization with complex constraints," *Inf. Sci.*, vol. 560, pp. 68–91, Jun. 2021.
- [63] V. L. Huang, P. N. Suganthan, A. K. Qin, and S. Baskar, "Multiobjective differential evolution with external archive and harmonic distance-based diversity measure," Nanyang Technological University, Singapore, Tech. Rep., 2005.
- [64] A. Zhou, J. Sun, and Q. Zhang, "An estimation of distribution algorithm with cheap and expensive local search methods," *IEEE Trans. Evol. Comput.*, vol. 19, no. 6, pp. 807–822, Dec. 2015.
- [65] Z.-J. Wang, Y.-R. Zhou, and J. Zhang, "Adaptive estimation distribution distributed differential evolution for multimodal optimization problems," *IEEE Trans. Cybern.*, vol. 52, no. 7, pp. 6059–6070, Jul. 2022.
- [66] C. A. C. Coello, G. B. Lamont, and D. A. Van Veldhuizen, *Evolutionary Algorithms for Solving Multi-Objective Problems*. 2nd ed. New York, USA: Springer, 2007.
- [67] A. Zhou, Q. Zhang, and Y. Jin, "Approximating the set of Pareto-optimal solutions in both the decision and objective spaces by an estimation of distribution algorithm," *IEEE Trans. Evol. Comput.*, vol. 13, no. 5, pp. 1167–1189, Oct. 2009.
- [68] E. Zitzler and L. Thiele, "Multiobjective evolutionary algorithms: A comparative case study and the strength Pareto approach," *IEEE Trans. Evol. Comput.*, vol. 3, no. 4, pp. 257–271, Nov. 1999.
- [69] W. Li, T. Zhang, R. Wang, S. Huang, and J. Liang, "Multimodal multi-

objective optimization: Comparative study of the state-of-the-art,” *Swarm Evol. Comput.*, vol. 77, p. 101253, Mar. 2023.

- [70] M. Mitchell, *An Introduction to Genetic Algorithms*. Cambridge, UK: MIT Press, 1998.
- [71] L. van der Maaten and G. Hinton, “Visualizing data using t-SNE,” *J. Mach. Learn. Res.*, vol. 9, no. 86, pp. 2579–2605, Nov. 2008.



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